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# Individual Assessment Coversheet

To be attached to the front of the assessment.

| | |
|---|---|
| **Campus:** | Pretoria |
| **Faculty:** | Information Systems |
| **Module Code:** | ITVNA0-B22 |
| **Group:** | Group 2 |
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| Indicate | Yes | No |
|---|---|---|
| Plagiarism report attached | Y | |

**Declaration:**

I declare that this assessment is my own original work except for source material explicitly acknowledged. I also declare that this assessment or any other of my original work related to it has not been previously, or is not being simultaneously, submitted for this or any other course. I am aware of the AI policy and acknowledge that I have not used any AI technology to generate or manipulate data, other than as permitted by the assessment instructions. I also declare that I am aware of the Institution's policy and regulations on honesty in academic work as set out in the Conditions of Enrolment, and of the disciplinary guidelines applicable to breaches of such policy and regulations.

| Signature | Date |
|---|---|
| A Makhado | 01/06/2026 |

**Lecturer's Comments:**

| |
|---|
| |

| Marks Awarded: | % |
|---|---|
| | |

| Signature | Date |
|---|---|
| | |

# ACKNOWLEDGEMENT

I would like to formally acknowledge the invaluable support, guidance, and contributions that made the successful completion of this project possible.

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I would also like to extend my gratitude to Eduvos for providing a conducive learning environment, access to educational resources, and the technological tools necessary to complete this assignment. The institution's commitment to academic excellence has greatly contributed to my growth and understanding of networking concepts and their practical application in modern IT environments.

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This acknowledgement stands as a reflection of the collective effort, support, and dedication of all those who contributed to the successful completion of this project.

# Abstract

**ABSTRACT**

The rapid growth of cloud computing, mobile learning, and smart campus technologies has increased the demand for scalable, secure, and efficient network infrastructures in higher education institutions. Eduvos is preparing to modernise its digital environment by adopting IPv6, implementing Virtual Local Area Networks (VLANs), deploying Dynamic Host Configuration Protocol (DHCP), and redesigning network addressing through subnetting. These technologies aim to improve network performance, security, scalability, and management efficiency across multiple campuses.

This report evaluates the role of IPv6 in supporting cloud integration, mobile device connectivity, and emerging technologies such as the Internet of Things (IoT). It further examines the challenges associated with IPv4-to-IPv6 migration and the long-term benefits of IPv6 adoption. The report also demonstrates the implementation of VLAN segmentation to separate staff and student network traffic, enhancing both security and bandwidth utilisation. Additionally, the report explores the use of DHCP to automate IP address allocation, reduce configuration errors, and improve administrative efficiency. Finally, a subnetting design is presented for the Eduvos Tygervalley Campus, showing how a single network can be divided into multiple departmental subnets while maintaining communication through routing technologies.

The findings highlight the importance of modern networking solutions in creating a reliable, scalable, and future-ready educational environment capable of supporting increasing numbers of users, devices, and digital services.

# Executive Summary

Eduvos is undergoing digital transformation to support cloud-based learning platforms, remote access services, smart campus technologies, and a growing number of connected devices. To achieve these objectives, the institution must modernise its network infrastructure by implementing advanced networking technologies that improve scalability, performance, security, and manageability.

This report focuses on four key networking areas. Firstly, it examines the adoption of IPv6 as a solution to address IPv4 limitations and support cloud computing, mobile connectivity, and future technology expansion. The report discusses both the advantages of IPv6 and the challenges associated with migration from existing IPv4 environments.

Secondly, the report investigates the use of Virtual Local Area Networks (VLANs) to separate staff and student traffic within the Eduvos Pretoria Campus network. VLAN segmentation reduces unnecessary network traffic, enhances performance, and provides additional security by isolating sensitive resources.

Thirdly, the report evaluates the implementation of Dynamic Host Configuration Protocol (DHCP) to automate IP address allocation in student computer laboratories. DHCP reduces administrative workload, eliminates common configuration errors, and ensures consistent network connectivity for users.

Lastly, the report presents a subnetting strategy for Eduvos Tygervalley Campus by dividing a single Class C network into four equal subnets for different departments. The design demonstrates efficient IP address utilisation and explains how routing technologies facilitate communication between departments when required.

Collectively, these networking solutions provide Eduvos with a modern, secure, and scalable infrastructure capable of supporting current educational requirements and future technological advancements.

# Table of Contents


**ACKNOWLEDGEMENT** .......... 2

**Abstract** .......... 3

**Executive Summary** .......... 4

**Introduction** .......... 8

**Project Resources** .......... 9

Section 1: Virtualisation Software Installation .......... 9

Step 1: Download and Install Oracle VirtualBox .......... 9

Section 1.1: Operating System Preparation – ISO Download .......... 10

Step 1.1: Downloading Windows 11 ISO .......... 10

Section 2: Virtual Machine Creation .......... 10

Step 2: Starting a New Virtual Machine .......... 10

Section 3: Virtual Machine Setup – Windows 11 .......... 11

Step 3: Configuring VM 01 .......... 11

Section 4: Unattended Installation Setup – VM 01 .......... 12

Step 4: Configuring Installation Parameters .......... 12

Section 5: Resource Allocation – VM 01 .......... 14

Step 5: Configuring Virtual Hardware .......... 14

Section 6: Storage Configuration – VM 01 .......... 15

Step 6: Configuring Virtual Hard Disk .......... 15

Section 7: VM Creation Completed .......... 17

Step 7: Finalizing VM Setup .......... 17

Section 8: Operating System Installation – VM 1 .......... 19

Step 8: Booting the Virtual Machine .......... 19

Section 9: Operating System Installation – VM 1 .......... 20

Step 9: Installing Windows 11 .......... 20

Section 11: Operating System Installation – VM 1 .......... 21

Step 9: Windows Login Screen .......... 21

Section 10: User Account Integration – VM 1 .......... 23

Step 10: Logging in and Linking Student Account .......... 23

Section 11: System Configuration – Firewall Setup ....................25
Step 11: Accessing Windows Defender Firewall ....................25
Section 12: Firewall Status – VM 1 ....................26
Step 12: Viewing Firewall Settings ....................26
Section 13: Firewall Rule Configuration – VM 1 ....................28
Step 13: Accessing Advanced Firewall Settings ....................28
Section 14: Custom Firewall Rule Management – VM 1 ....................30
Step 14: Reviewing Inbound Rules ....................30
Section 15: Custom Firewall Rule Creation – VM 1 ....................31
Step 15: Creating a New Inbound Rule ....................31
Section 16: Disk Management – Shrink Volume ....................33
Step 16: Shrinking the Windows (C:) Partition ....................33
Section 17: Partition Adjustment – Shrink Volume ....................35
Step 17: Shrinking C: Drive ....................35
Section 18: Command Line Access – VM 1 ....................36
Step 18: Opening Command Prompt ....................36
Section 19: Command Line Administration – Network Test ....................36
Step 19: Checking Network Settings ....................36
**Question 1** ....................38
**Question 2** ....................39
Starting Point: Installing Cisco Packet Tracer ....................40
Locating Cisco Packet Tracer After Installation ....................41
Accessing Cisco Packet Tracer After Installation ....................42
2.1. Create two VLANs on a managed switch: ....................43
2.2. Assign Client Machines to VLANs ....................45
2.2. Assign Client Machines to VLANs ....................46
**Question 3** ....................50
3.1. In your own words, describe the role of DHCP in a network. ....................51
3.2. Explain how DHCP improves efficiency compared to manual IP configuration. ....................52
3.3. Design a DHCP Scope ....................53
3.4 How a Client Requests and Receives an IP Address from DHCP ....................56

3.5. Summarise the benefits Eduvos gains from DHCP automation. ........................59

**Question 4** ........................................................................................60

4.1 Subnetting Calculations ....................................................................62

4.2 Assign one subnet to each department ...............................................67

4.3. Explain how a router (or Layer 3 switch) would enable communication between the four subnets. ..............................................................................71

4.4. Describe how you would verify connectivity using tools such as ping or tracert. Provide a simple diagram showing the router connecting the four subnets. .............73

**CONCLUSION** ......................................................................................75

**Referencing** ........................................................................................76

# Introduction

Modern educational institutions rely heavily on digital technologies to deliver teaching, learning, administration, and communication services. As the number of users, devices, and online applications continues to increase, network infrastructures must evolve to meet growing demands for performance, reliability, security, and scalability. Eduvos is currently enhancing its technological environment through the introduction of cloud-connected learning platforms, smart campus systems, and remote learning capabilities.

To support these developments, several networking technologies play a critical role. Internet Protocol Version 6 (IPv6) provides a significantly larger address space than IPv4, enabling institutions to accommodate growing numbers of devices while improving connectivity and network efficiency. Virtual Local Area Networks (VLANs) allow network administrators to logically separate users and resources, improving both performance and security. Dynamic Host Configuration Protocol (DHCP) automates the allocation of IP addresses and network settings, reducing administrative effort and minimising configuration errors. In addition, subnetting enables organisations to divide larger networks into smaller, more manageable segments, improving traffic management and network control.

This report investigates how these technologies can be applied within Eduvos to support its digital transformation objectives. The report evaluates IPv6 adoption, VLAN implementation, DHCP deployment, and subnetting design while demonstrating how each technology contributes to a secure, efficient, and future-ready network infrastructure. Through practical networking concepts and theoretical designs, the report highlights the importance of modern network management practices in supporting the institution's operational and educational goals.

# Project Resources

## Section 1: Virtualisation Software Installation

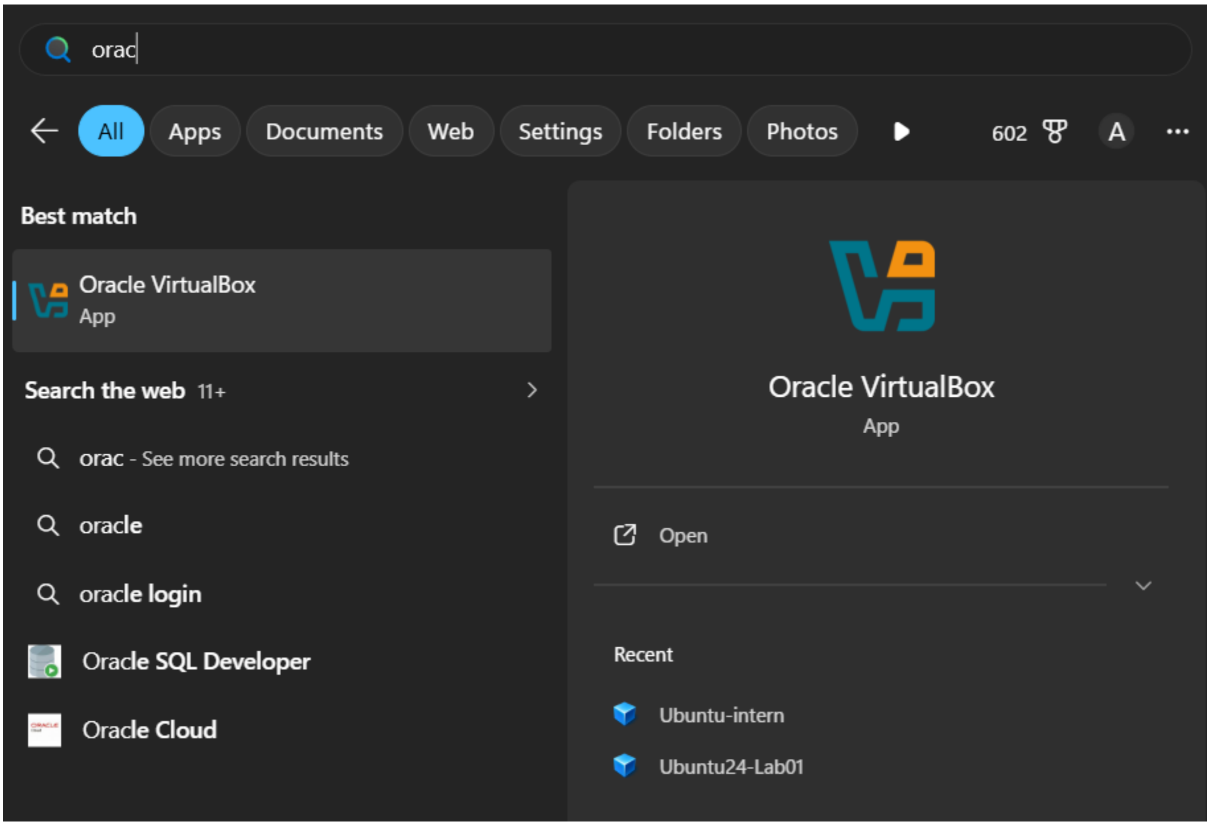
### Step 1: Download and Install Oracle VirtualBox

I have downloaded and installed Oracle VirtualBox, which is an extremely popular open-source virtual machine application. This application allows users to create and run many different kinds of VMs on just one host computer, using either Windows or Linux OS.

I chose this software because it supports the objectives outlined in CompTIA A+ 220 1202 and offers a reliable area for performing tests and configuration jobs.

**Evidence:**

- The screenshot shows Oracle VirtualBox successfully installed and accessible from the Windows search interface.




**Evaluation:**

- Installing VirtualBox is a critical prerequisite for the project, as it provides the virtualization layer required to run both Windows and Linux VMs.
- This step ensures the host machine (Windows 10/11) is properly equipped to support the project's technical objectives.

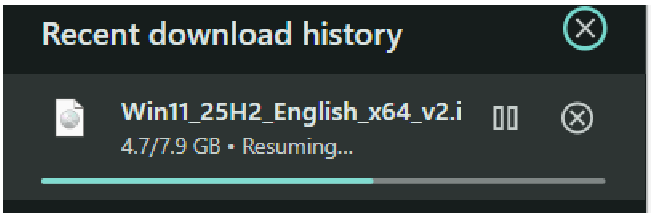
---

## Section 1.1: Operating System Preparation – ISO Download

### Step 1.1: Downloading Windows 11 ISO

- Before installing Windows 11 in the VM, I downloaded the official installation file.
- The file name was **Win11_25H2_English_x64_v2.iso**.
- The download progress showed **4.7 GB of 7.9 GB** completed, confirming the ISO was being retrieved successfully.

**Evidence :**




- The download window shows the ISO file in progress.
- This proves the operating system image was obtained and prepared for use in VirtualBox.

**Evaluation:**

- Downloading the ISO is a critical prerequisite for VM setup.
- It ensures the VM can boot and install Windows 11 properly.
- This step validates that the environment was prepared correctly before installation.

---

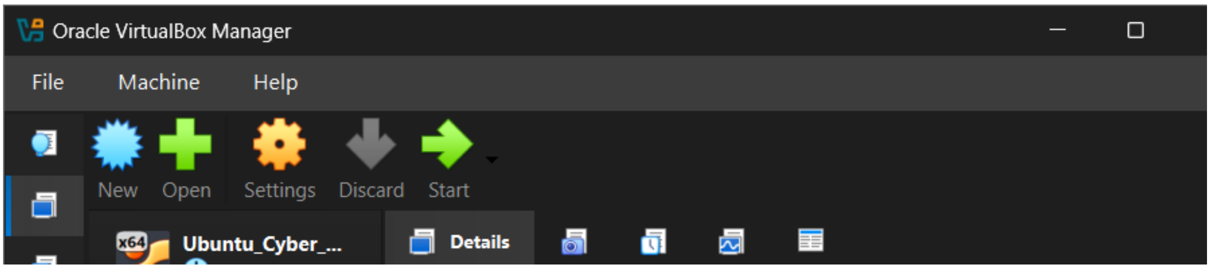
# Section 2: Virtual Machine Creation

### Step 2: Starting a New Virtual Machine

- In Oracle VirtualBox Manager, I pressed the **blue "New" button** to begin creating a new virtual machine.

- This initiated the VM setup wizard, allowing me to specify the name, operating system type, and hardware resources for the new environment.

**Evidence :**




- This confirms that the VM creation process was successfully initiated and saved.

**Evaluation:**

- Creating VM 1 establishes the Windows environment required by the project brief.
- This step demonstrates the ability to allocate system resources (CPU, RAM, disk space) and configure essential components such as storage controllers and network adapters.
- VM creation is a critical milestone because it provides the isolated environment where all Windows-based configuration tasks will be performed.

---

# Section 3: Virtual Machine Setup – Windows 11

## Step 3: Configuring VM 01

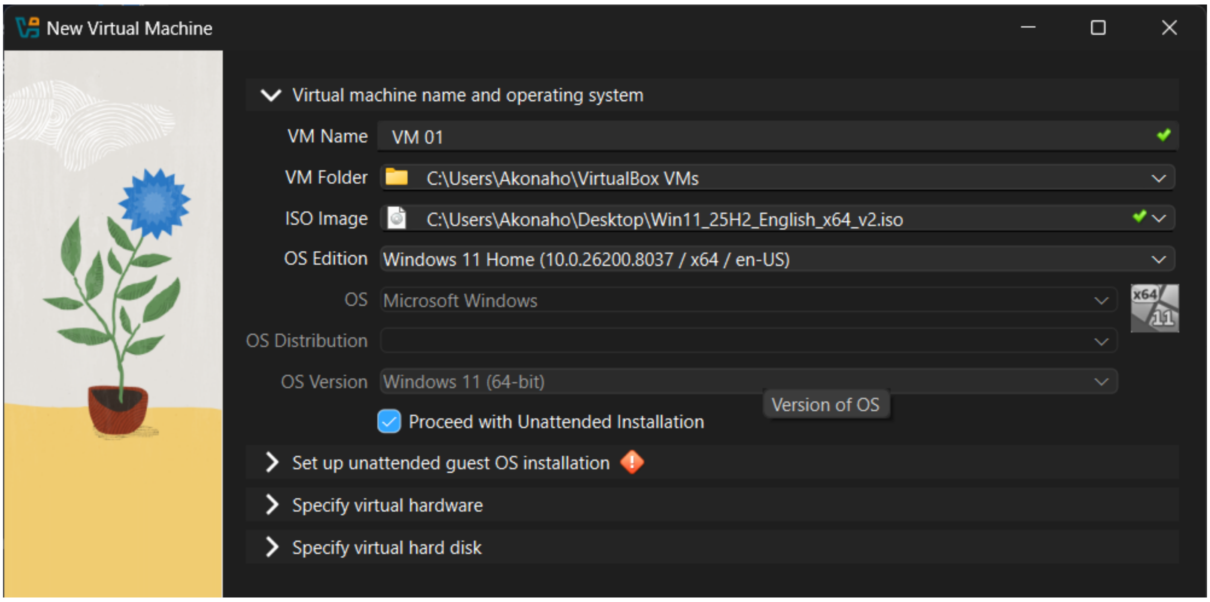
- I created a new virtual machine named **"VM 01"**.
- The VM folder was set to: C:\Users\Akonaho\VirtualBox VMs — this ensures all VM files are stored in a dedicated directory for easy management.
- The installation ISO selected was: Win11_25H2_English_x64_v2.iso, which contains the Windows 11 Home edition installer.
- Operating System details:
  - **OS Edition:** Windows 11 Home (10.0.26200.8037 / x64 / en-US)
  - **OS Type:** Microsoft Windows

- **OS Version:** Windows 11 (64-bit)

**Unattended Installation:**

- The option **"Proceed with Unattended Installation"** was enabled.
- This feature allows VirtualBox to automatically handle installation prompts (like username, password, product key, and regional settings), speeding up the setup process.

**Evidence:**




- The wizard clearly shows the VM name, ISO path, OS edition, and unattended installation option selected.
- This confirms that the VM is properly configured to begin the Windows 11 installation process.

**Evaluation:**

- Choosing Windows 11 Home aligns with the project requirement to create a Windows VM.
- Using an ISO ensures a clean installation environment.
- Enabling unattended installation demonstrates efficiency and automation in system setup, which is a key virtualization skill.

# Section 4: Unattended Installation Setup – VM 01

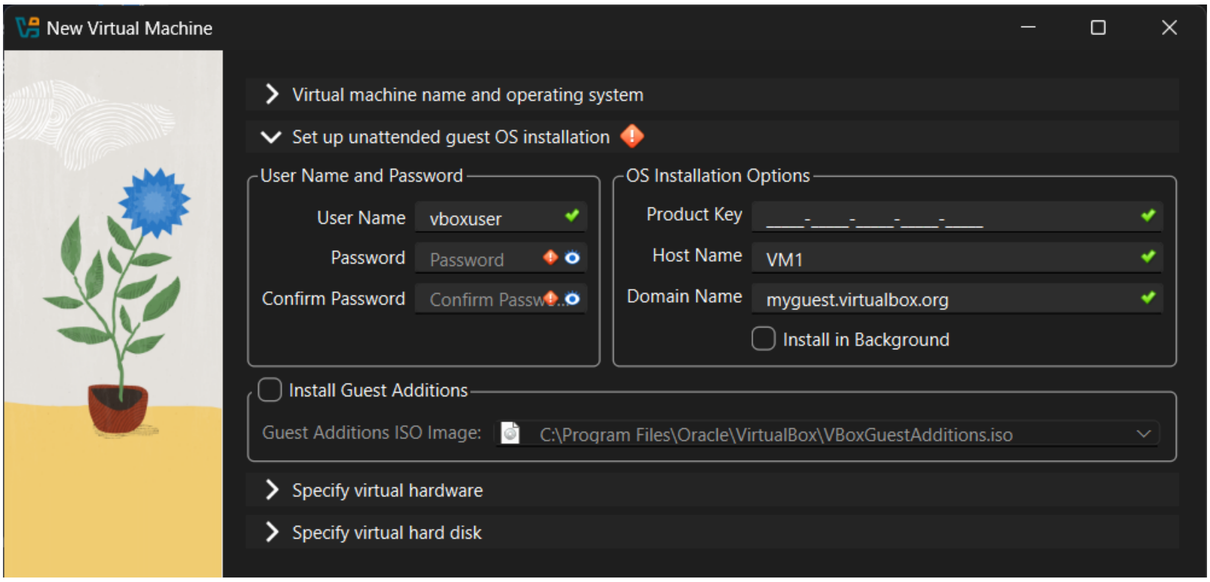
## Step 4: Configuring Installation Parameters

- During the VM creation process, I enabled **Unattended Guest OS Installation**.
- This feature allows VirtualBox to automatically complete the Windows installation without requiring manual input for each step.

**Configuration Details:**

- **User Credentials:**
  - Username: vboxuser
  - Password: Password (entered and confirmed)
- **System Identity:**
  - Host Name: VM1
  - Domain Name: myguest.virtualbox.org
- **Guest Additions:**
  - Selected ISO: VBoxGuestAdditions.iso
  - Purpose: Provides enhanced integration features such as shared clipboard, drag-and-drop, and improved display drivers.

**Evidence :**




- The wizard clearly shows the username, password, host name, and domain name fields filled in.

- The Guest Additions ISO path is specified, confirming that integration tools will be installed alongside the OS.

**Evaluation:**

- Automating the installation process saves time and ensures consistency across VM setups.
- Defining credentials and host/domain names demonstrates proper system configuration practices.
- Including Guest Additions enhances usability, making the VM more responsive and better integrated with the host system.

---

# Section 5: Resource Allocation – VM 01

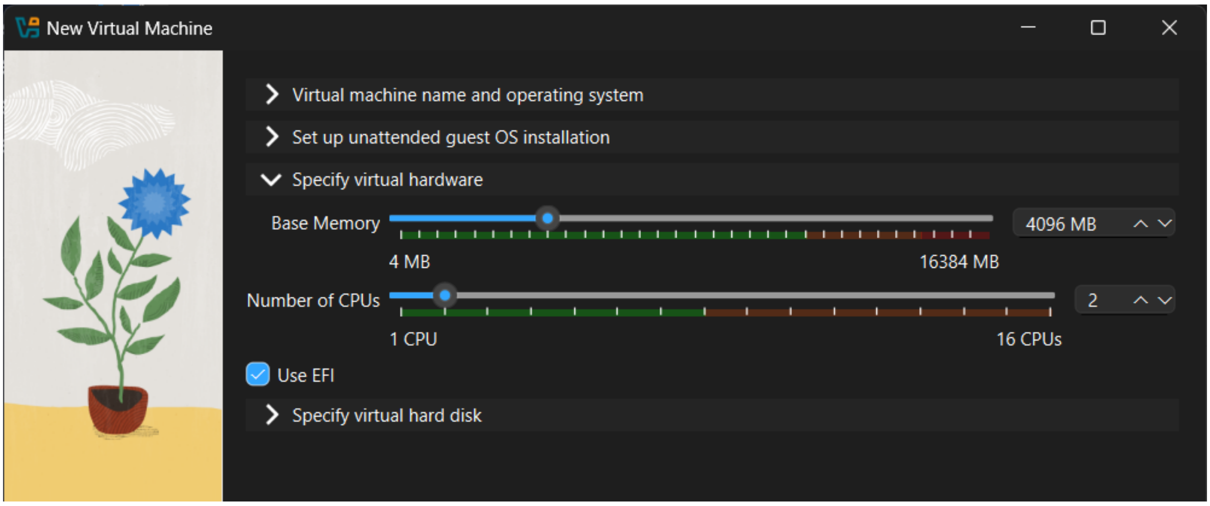
## Step 5: Configuring Virtual Hardware

- I specified the hardware resources for the new virtual machine to ensure stable performance while balancing host system capacity.

**Configuration Details:**

- **Base Memory (RAM):** 4096 MB (4 GB)
    - Allocating 4 GB ensures Windows 11 can run smoothly with basic applications, while leaving sufficient memory for the host system.
- **Number of CPUs:** 2
    - Assigning 2 virtual CPUs provides multitasking capability without overloading the host machine.
- **EFI Enabled:** Checked
    - EFI (Extensible Firmware Interface) is required for modern operating systems like Windows 11, ensuring compatibility with secure boot and advanced hardware features.
- **Virtual Hard Disk:** To be specified in the next step (e.g., 80 GB dynamically allocated).

**Evidence :**




- The wizard shows sliders set to 4096 MB RAM and 2 CPUs, with EFI enabled.
- This confirms that the VM has been provisioned with adequate resources for installation and operation.

**Evaluation:**

- Allocating resources is a critical step in virtualization, as it directly impacts VM performance.
- The chosen values balance efficiency and host system stability, demonstrating proper resource management.
- Enabling EFI ensures the VM is future-proof and aligned with modern OS requirements.

---

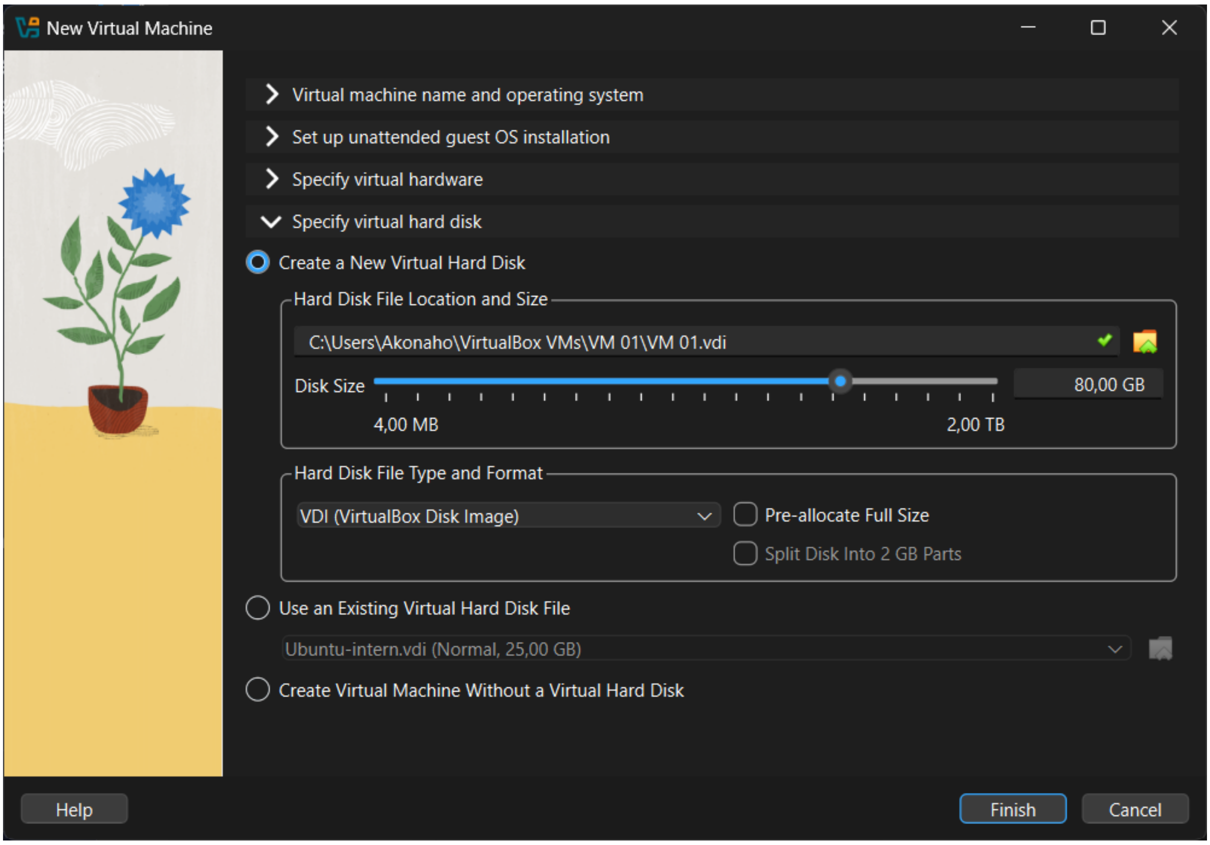
# Section 6: Storage Configuration – VM 01

## Step 6: Configuring Virtual Hard Disk

- I selected the option to **Create a New Virtual Hard Disk** for VM 01.
- **File Location:** C:\Users\Akonaho\VirtualBox VMs\VM 01\VM 01.vdi
- **Disk Size:** 80 GB (dynamically allocated)

    - This size is sufficient for Windows 11 installation, updates, and basic applications.
- **File Type:** VDI (VirtualBox Disk Image)
    - Chosen for compatibility and flexibility within VirtualBox.
- **Pre-allocate Full Size:** Not selected, allowing dynamic disk growth to save host storage space.
- **Split Disk Into 2 GB Parts:** Not selected, keeping the disk file unified for easier management.

**Evidence :**




- The wizard shows the disk size set to 80 GB, file type as VDI, and storage path correctly defined.
- This confirms the VM has a dedicated virtual hard disk ready for installation.

**Evaluation:**

- Allocating 80 GB balances performance and storage efficiency.
- Dynamic allocation ensures the VM only uses physical disk space as needed.

- Choosing VDI format aligns with VirtualBox's native disk management, simplifying backups and portability.

# Section 7: VM Creation Completed

## Step 7: Finalizing VM Setup

- After pressing **Finish**, the virtual machine **VM 01** was successfully created.
- VM 01 now appears in the VirtualBox Manager list, ready for operating system installation and configuration.

**Evaluation:**

- Completing this step establishes the Windows 11 VM environment required by the project brief.
- The VM is now prepared for practical tasks such as OS installation, configuration, and troubleshooting.

---

**Section 7: Completed Virtual Machine – VM 1 (Windows 11)**

**General Configuration:**

- **Name:** VM 1
- **Operating System:** Windows 11 (64-bit)

**Display Settings:**

- Video Memory: 128 MB
- Graphics Controller: VMSVGA
- Remote Desktop Server: Disabled
- Recording: Disabled

**Storage Setup:**

- Controller: SATA
- SATA Port 0: VM 1.vdi (Normal, 80 GB)

- SATA Port 1: Optical Drive (Windows 11 ISO attached for installation)

**Audio Settings:**

- Host Driver: Default
- Controller: Intel HD Audio

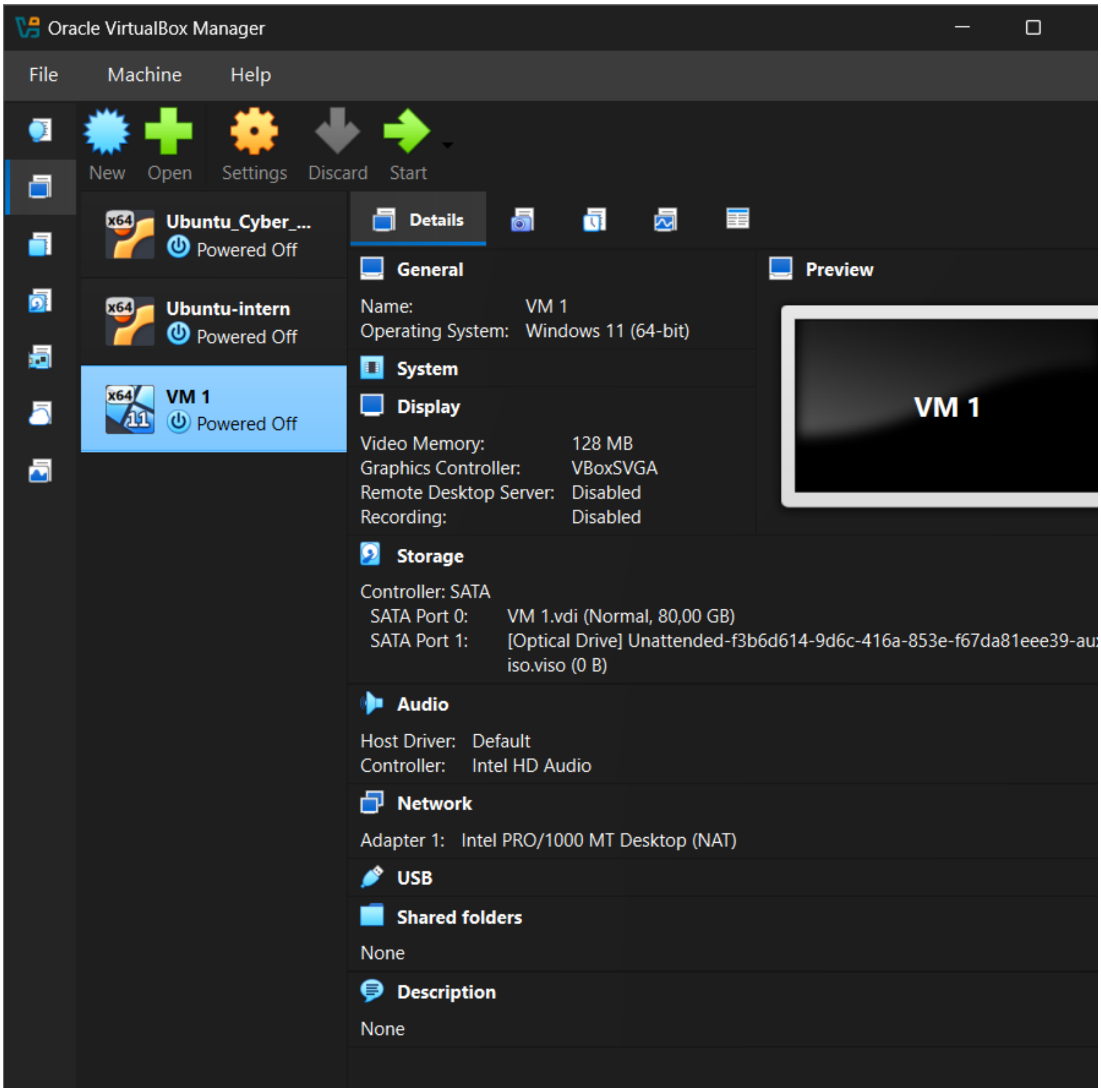
**Network Configuration:**

- Adapter 1: Intel PRO/1000 MT Desktop (NAT)
    - NAT mode allows the VM to access the internet through the host system while remaining isolated for security.

**Other Settings:**

- USB: None
- Shared Folders: None
- Description: None

**Evidence :**




- The VirtualBox Manager shows VM 1 listed and powered off, with all configuration details displayed in the right panel.
- The storage section confirms the virtual hard disk (80 GB VDI) and Windows 11 ISO are correctly attached, meaning the VM is ready for OS installation.

**Evaluation:**

- VM 1 has been successfully created and configured to meet the project requirement of a Windows virtual machine.
- The chosen settings (RAM, CPU, disk size, NAT networking) balance performance and host system stability.
- This VM is now prepared for the next stage: **Windows 11 installation and configuration tasks**

# Section 8: Operating System Installation – VM 1

## Step 8: Booting the Virtual Machine

- After pressing on **VM 1** in VirtualBox Manager, the system begins the startup process.
- The VirtualBox loading screen appears, confirming that the virtual machine is initializing and preparing to boot from the attached Windows 11 ISO.

**Evidence :**


- The VirtualBox logo and loading indicator are displayed, showing that VM 1 has successfully entered the boot sequence.
- This verifies that the VM is correctly linked to the installation media and ready to proceed with the Windows 11 setup.

**Evaluation:**

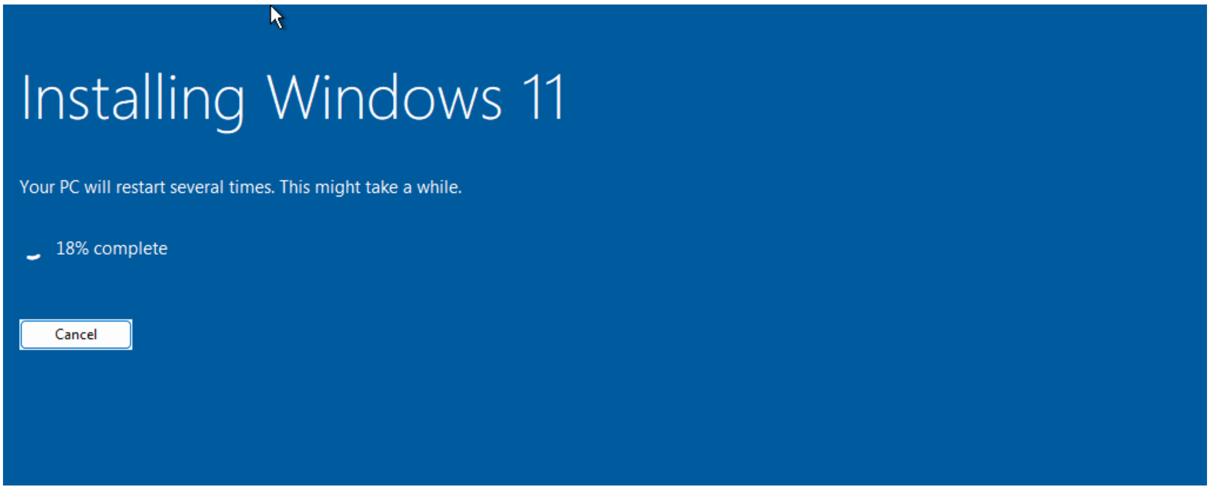
- Booting the VM is a critical step, as it transitions from configuration to actual operating system installation.
- The successful startup demonstrates that the hardware resources (RAM, CPU, disk, graphics controller) and storage configuration (ISO + VDI) were correctly defined during setup.
- This stage confirms readiness for the next phase: **Windows installation wizard and unattended setup execution**

# Section 9: Operating System Installation – VM 1

## Step 9: Installing Windows 11

- After starting VM 1, the system booted from the attached Windows 11 ISO and began the installation process.
- The installation wizard displayed progress with the message: *"Installing Windows 11 – Your PC will restart several times. This might take a while."*
- At this stage, the installation was **18% complete**, confirming that the OS setup was successfully underway.

**Evidence :**




- The installation progress screen verifies that Windows 11 is being installed on VM 1.
- The restart notifications indicate that the unattended installation process is functioning as expected.

**Evaluation:**

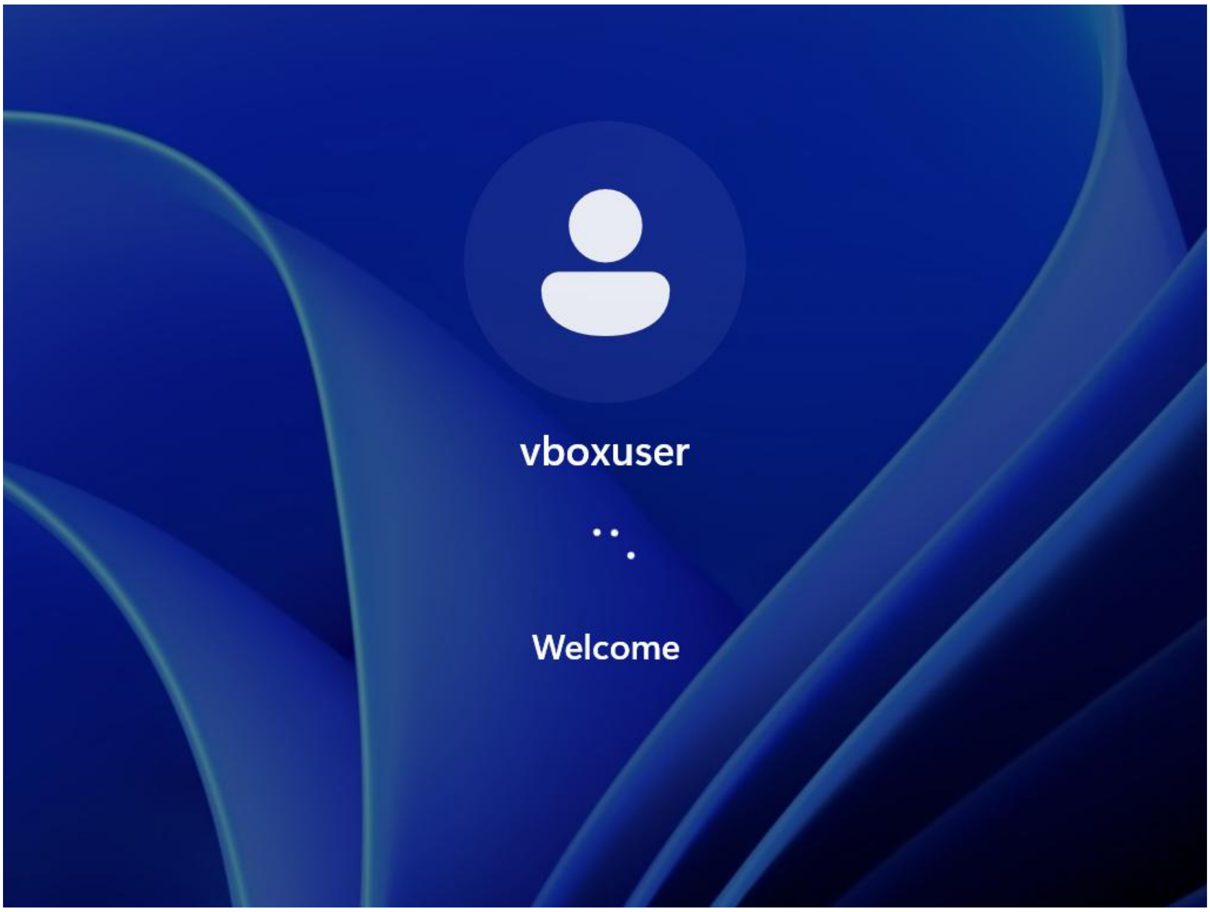
- This step demonstrates that the VM is properly configured to boot from the ISO and install the operating system.
- The progress bar confirms that the installation is active, validating earlier configuration choices (RAM, CPU, disk size, EFI).
- Once completed, the VM transitions to the login stage, where the default user (vboxuser) is created.

# Section 11: Operating System Installation – VM 1

## Step 9: Windows Login Screen

- After the installation process completed, VM 1 booted into the **Windows 11 login interface**.
- The system displayed the default user account created during unattended installation: **vboxuser**.
- The "Welcome" message confirms that the operating system has successfully initialized and is ready for first-time use.

**Evidence :**




- The login screen shows the user account vboxuser with a welcome prompt.
- This verifies that the unattended installation was successful and the VM is now running Windows 11.

**Evaluation:**

- Reaching the login screen demonstrates that the VM creation, resource allocation, and installation steps were correctly executed.
- The presence of the vboxuser account confirms that the unattended setup applied the credentials defined earlier in the configuration wizard.
- This stage marks the transition from installation to **system configuration and practical tasks** (e.g., disk management, firewall rules, command execution).

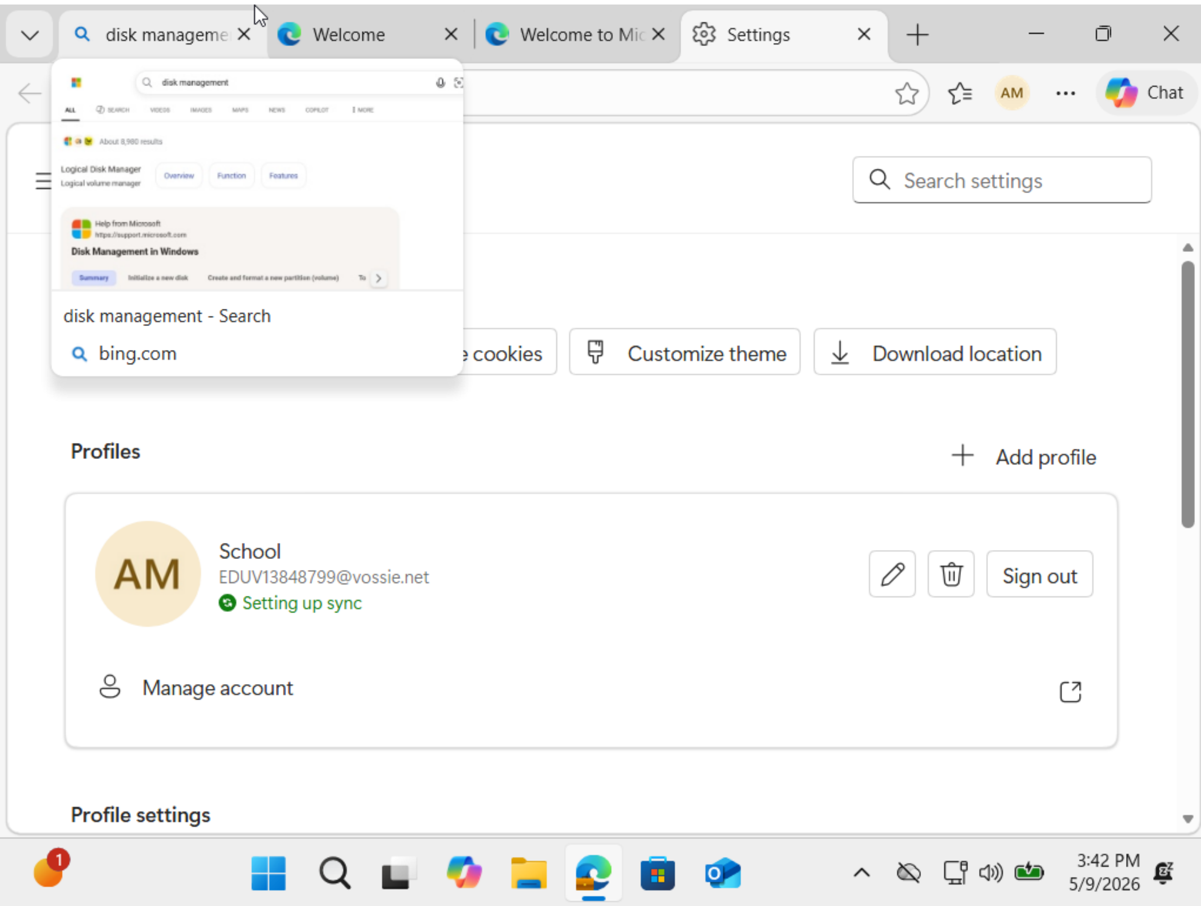
---

# Section 10: User Account Integration – VM 1

## Step 10: Logging in and Linking Student Account

- After the Windows 11 installation completed, I logged in using the default account created during unattended setup: **vboxuser**.
- Once inside the system, I navigated to **Settings → Accounts → Profiles**.
- I then signed in with my **student account (Eduvos email)** to ensure synchronization of settings, access to Microsoft services, and proper personalization under my academic identity.

**Evidence :**




- The Settings window shows the profile section with the student account (EDUV13848799@vossie.net) being set up for sync.
- This confirms that the VM is now linked to my student credentials, ensuring the environment reflects my actual user identity rather than the default vboxuser.

**Evaluation:**

- Linking the student account ensures that the VM can run properly with personalized settings, access to cloud services, and integration with academic resources.
- This step demonstrates good practice in system setup: moving from a generic installation account to a real user profile for practical use.
- It also aligns with the project requirement to show configuration tasks performed inside the VM environment.

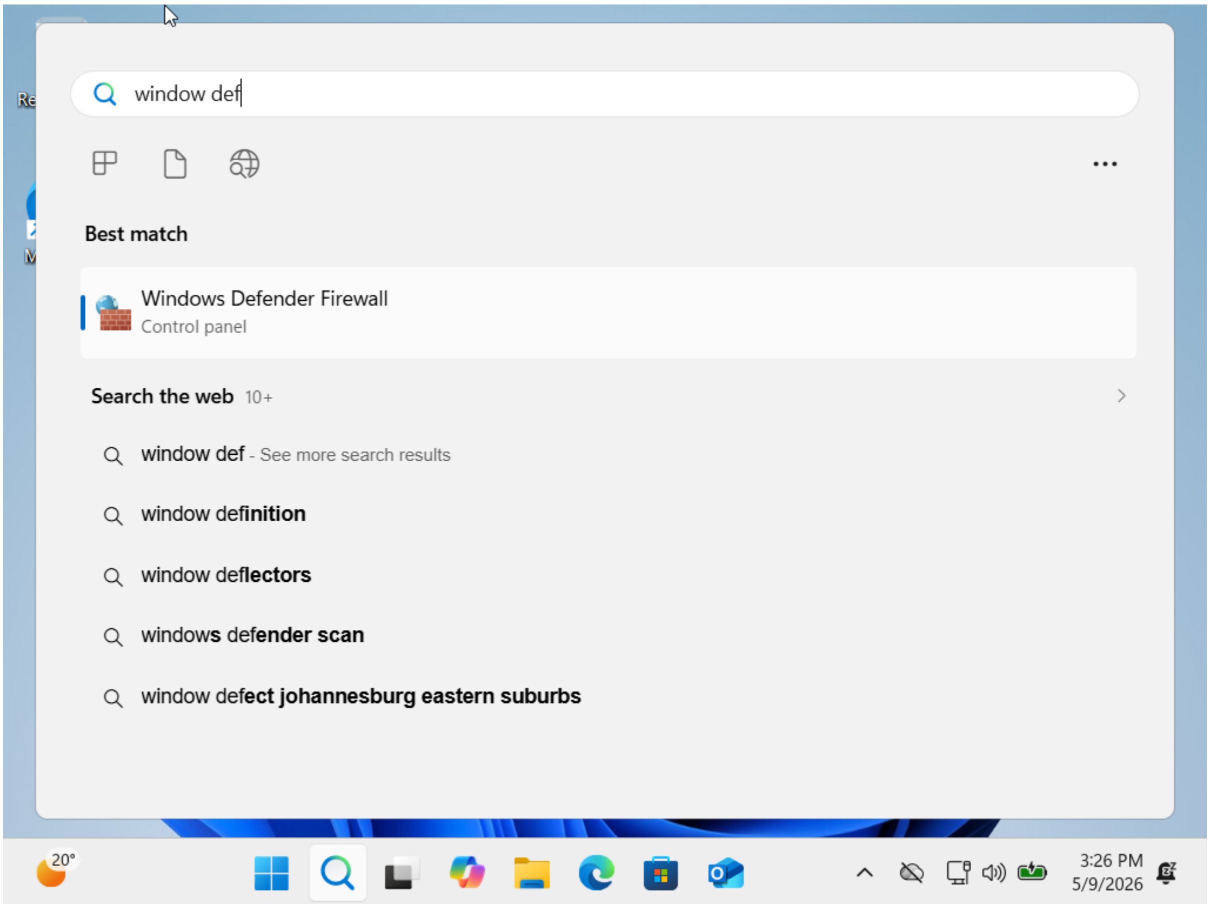
---

# Section 11: System Configuration – Firewall Setup

## Step 11: Accessing Windows Defender Firewall

- After logging in with my student account, I opened the **Start menu search bar** and typed **"window def"**.
- The best match result displayed was **Windows Defender Firewall (Control Panel)**.
- Selecting this option allows me to configure firewall rules, which are essential for securing the VM environment and controlling network traffic.

**Evidence :**




- The search interface shows "Windows Defender Firewall" as the best match.
- This confirms that the VM is running Windows 11 properly and that system security tools are accessible within the virtual environment.

**Evaluation:**

- Accessing Windows Defender Firewall is a critical configuration task, as it demonstrates the ability to manage system security inside the VM.
- This step aligns with the project requirement to perform practical tasks (like firewall rule creation) and provide screenshots as evidence.
- It also shows that the VM is functioning with full Windows features, ready for further configuration and troubleshooting.

---

# Section 12: Firewall Status – VM 1

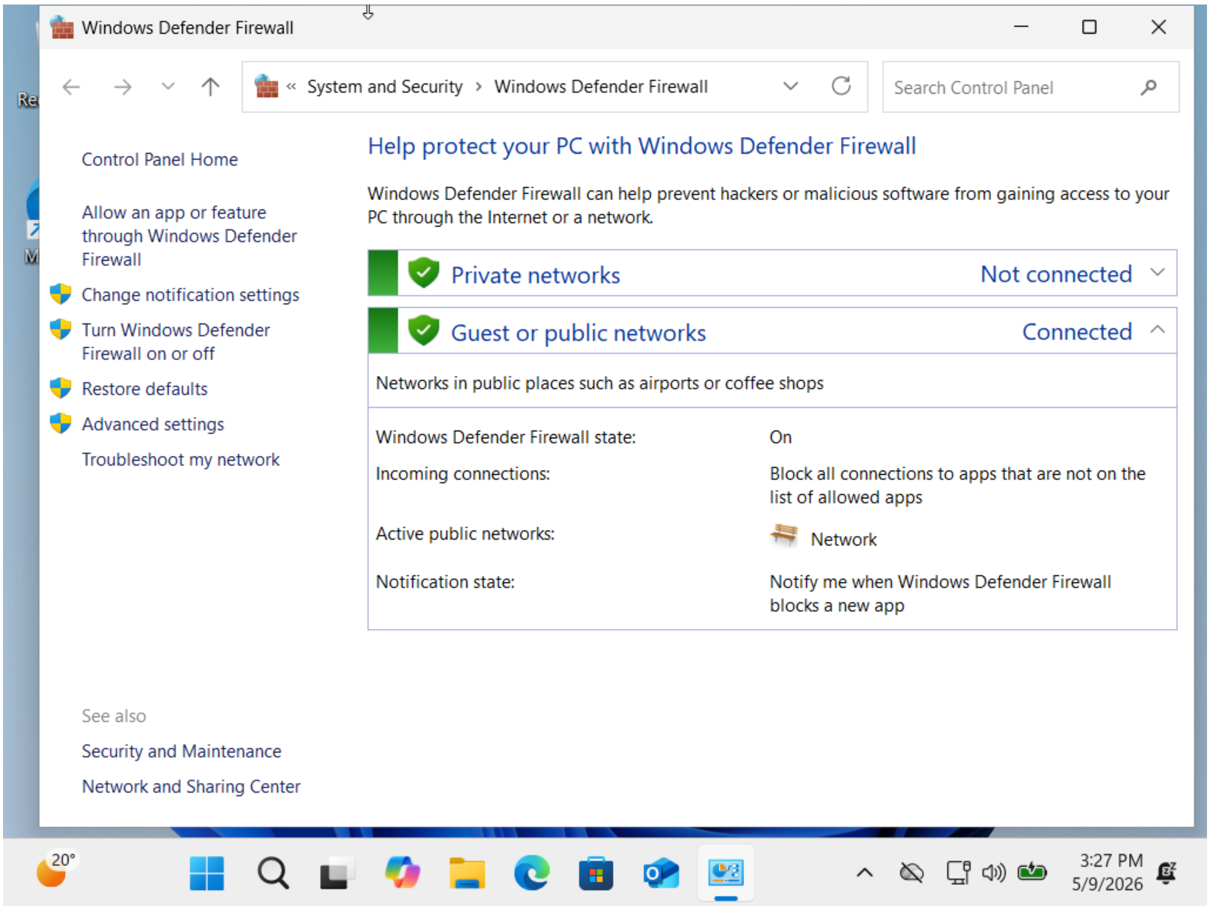
## Step 12: Viewing Firewall Settings

- After searching for Windows Defender Firewall, I opened the **Firewall control panel** to review the current security status.
- The system displayed firewall information for both **Private networks** and **Guest/Public networks**.
- The active connection was identified as a **public network**, with the firewall turned **On**.

**Configuration Details:**

- **Private Networks:** Not connected
- **Guest/Public Networks:** Connected
    - Firewall State: On
    - Incoming Connections: Block all apps not on the allowed list
    - Active Public Network: Network
    - Notification State: Notify when a new app is blocked

**Evidence :**




- The firewall window shows the active public network with protection enabled.
- This confirms that Windows Defender Firewall is running correctly inside VM 1 and is actively securing the environment.

**Evaluation:**

- Reviewing firewall status is a critical system configuration task, demonstrating awareness of network security within the VM.
- The settings confirm that the VM is protected against unauthorized inbound connections, aligning with best practices for secure system setup.
- This step provides clear evidence of security configuration, fulfilling part of the project's marking criteria.

# Section 13: Firewall Rule Configuration – VM 1

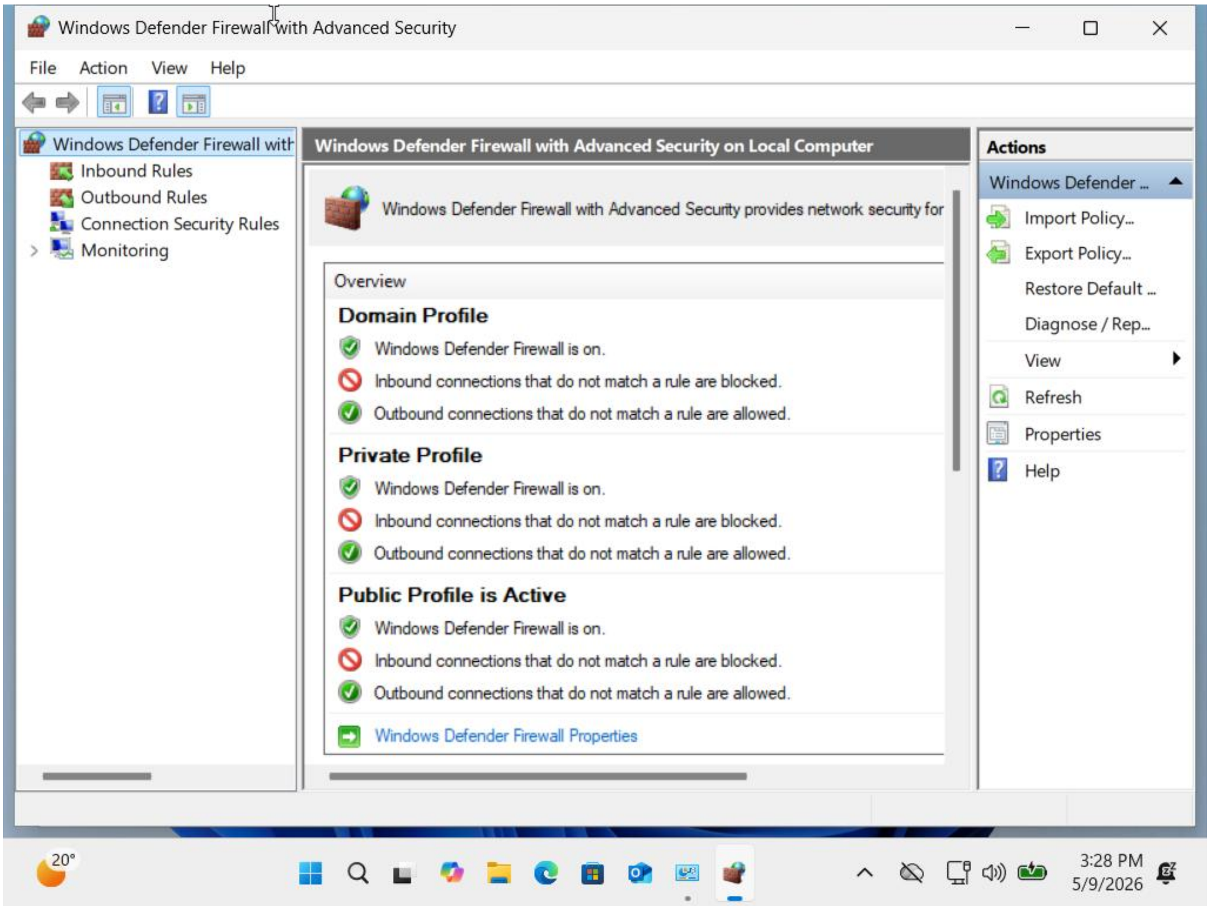
## Step 13: Accessing Advanced Firewall Settings

- From the Windows Defender Firewall control panel, I selected **Advanced Settings** to open the **Windows Defender Firewall with Advanced Security** console.
- This interface provides detailed control over inbound and outbound rules, connection security, and monitoring.

**Configuration Overview:**

- **Domain Profile:**
    - Firewall: On
    - Inbound connections not matching a rule: Blocked
    - Outbound connections not matching a rule: Allowed
- **Private Profile:**
    - Firewall: On
    - Inbound connections not matching a rule: Blocked
    - Outbound connections not matching a rule: Allowed
- **Public Profile (Active):**
    - Firewall: On
    - Inbound connections not matching a rule: Blocked
    - Outbound connections not matching a rule: Allowed

**Evidence:**




- The console shows all three profiles with firewall protection enabled.
- The active profile is **Public**, confirming that the VM is connected to a public network and is secured against unauthorized inbound traffic.

**Evaluation:**

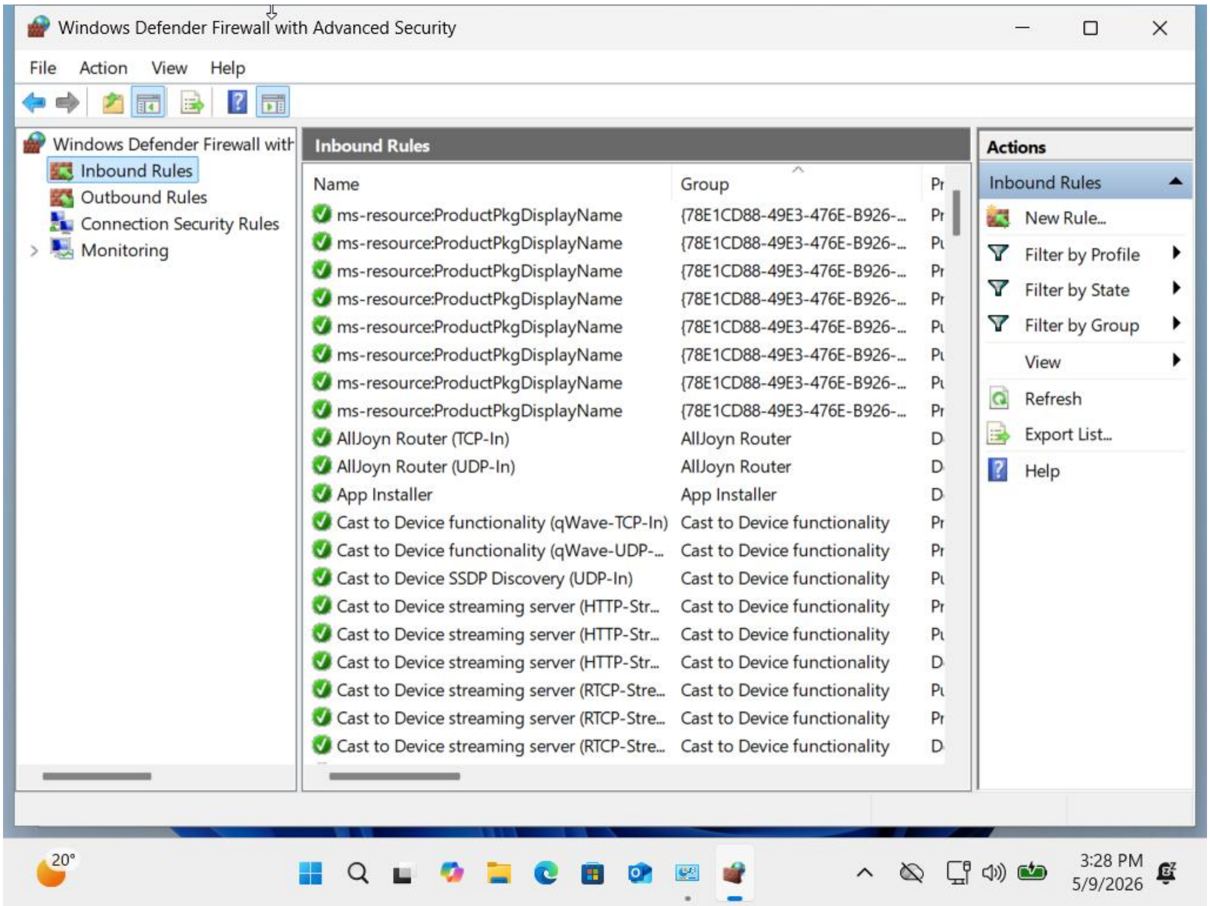
- Accessing advanced firewall settings demonstrates the ability to manage system security at a granular level.
- The configuration ensures that the VM is protected across all network profiles, with strict inbound rules and controlled outbound traffic.
- This step aligns with the project requirement to show practical configuration tasks and provide evidence through screenshots.

# Section 14: Custom Firewall Rule Management – VM 1

## Step 14: Reviewing Inbound Rules

- From the advanced firewall console, I navigated to **Inbound Rules** to examine how incoming network traffic is controlled.
- The list displayed multiple predefined rules, including entries for system services and applications such as **AllJoyn Router (TCP/UDP)**, **App Installer**, and **Cast to Device functionality**.
- Each rule specifies whether certain types of traffic are allowed or blocked, ensuring that only authorized connections reach the VM.

**Evidence :**




- The inbound rules list confirms that Windows Defender Firewall is actively managing incoming connections.
- The presence of multiple predefined rules demonstrates that the VM is protected by default configurations, which can be customized further.

**Evaluation:**

- Reviewing inbound rules is a critical step in system security management.
- It shows the ability to identify existing protections and prepare for custom rule creation (e.g., blocking a specific port or allowing a trusted application).
- This step aligns with the project requirement to demonstrate practical firewall configuration tasks inside the VM environment.

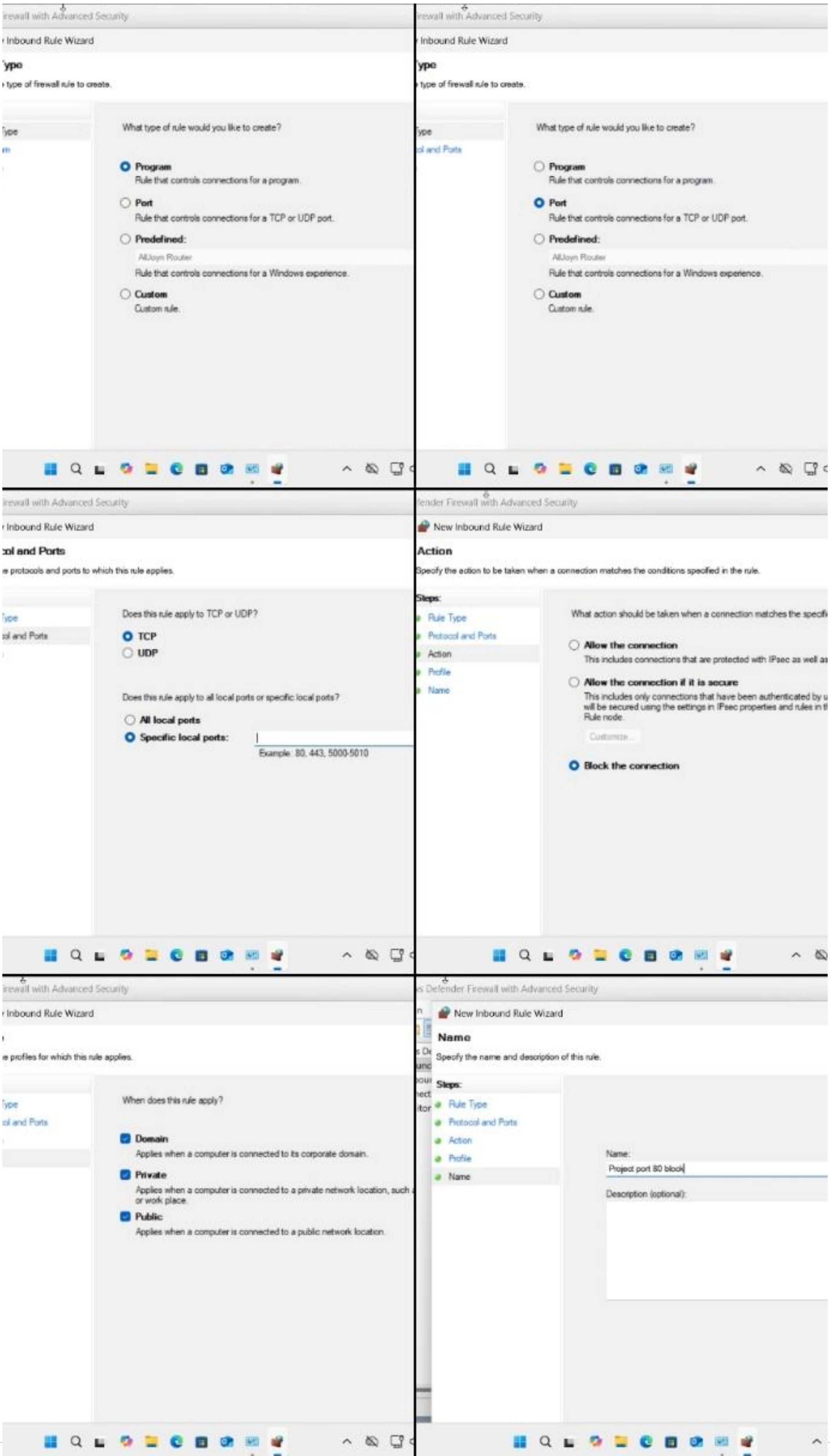
---

# Section 15: Custom Firewall Rule Creation – VM 1

## Step 15: Creating a New Inbound Rule

- I opened the **New Inbound Rule Wizard** in Windows Defender Firewall with Advanced Security.
- **Rule Type:** Selected **Port**, to control traffic through a specific TCP port.
- **Protocol and Ports:**
    - Protocol: TCP
    - Specific Local Port: **80** (commonly used for HTTP traffic)
- **Action:** Selected **Block the connection**, ensuring that any attempt to use port 80 is denied.
- **Profile:** Applied the rule to all profiles (Domain, Private, and Public) to ensure consistent protection across different network environments.
- **Name:** Entered **“Project port 80 block”** to clearly identify the rule.

**Evidence :**


New Inbound Rule Wizard
What type of rule would you like to create?
Program
Rule that controls connections for a program.
Port
Rule that controls connections for a TCP or UDP port.
Predefined:
AllJoyn Router
Rule that controls connections for a Windows experience.
Custom
Custom rule.
Protocol and Ports
Does this rule apply to TCP or UDP?
TCP
UDP
Does this rule apply to all local ports or specific local ports?
All local ports
Specific local ports:
Example: 80, 443, 5000-5010
Action
Specify the action to be taken when a connection matches the conditions specified in the rule.
Steps:
Rule Type
Protocol and Ports
Action
Profile
Name
Allow the connection
Allow the connection if it is secure
Customize...
Block the connection
When does this rule apply?
Domain
Applies when a computer is connected to its corporate domain.
Private
Public
Applies when a computer is connected to a public network location.
Name
Specify the name and description of this rule.
Name:
Project port 80 block
Description (optional):

- The wizard shows the rule creation steps completed, with the final name "Project port 80 block."
- This confirms that the inbound rule was successfully defined and saved.

**Evaluation:**

- Blocking port 80 demonstrates the ability to create custom firewall rules for enhanced security.
- This rule prevents unauthorized HTTP traffic from entering the VM, reducing exposure to potential web-based attacks.
- The step fulfils the project requirement to show practical firewall configuration and provides clear evidence of security management inside the VM.

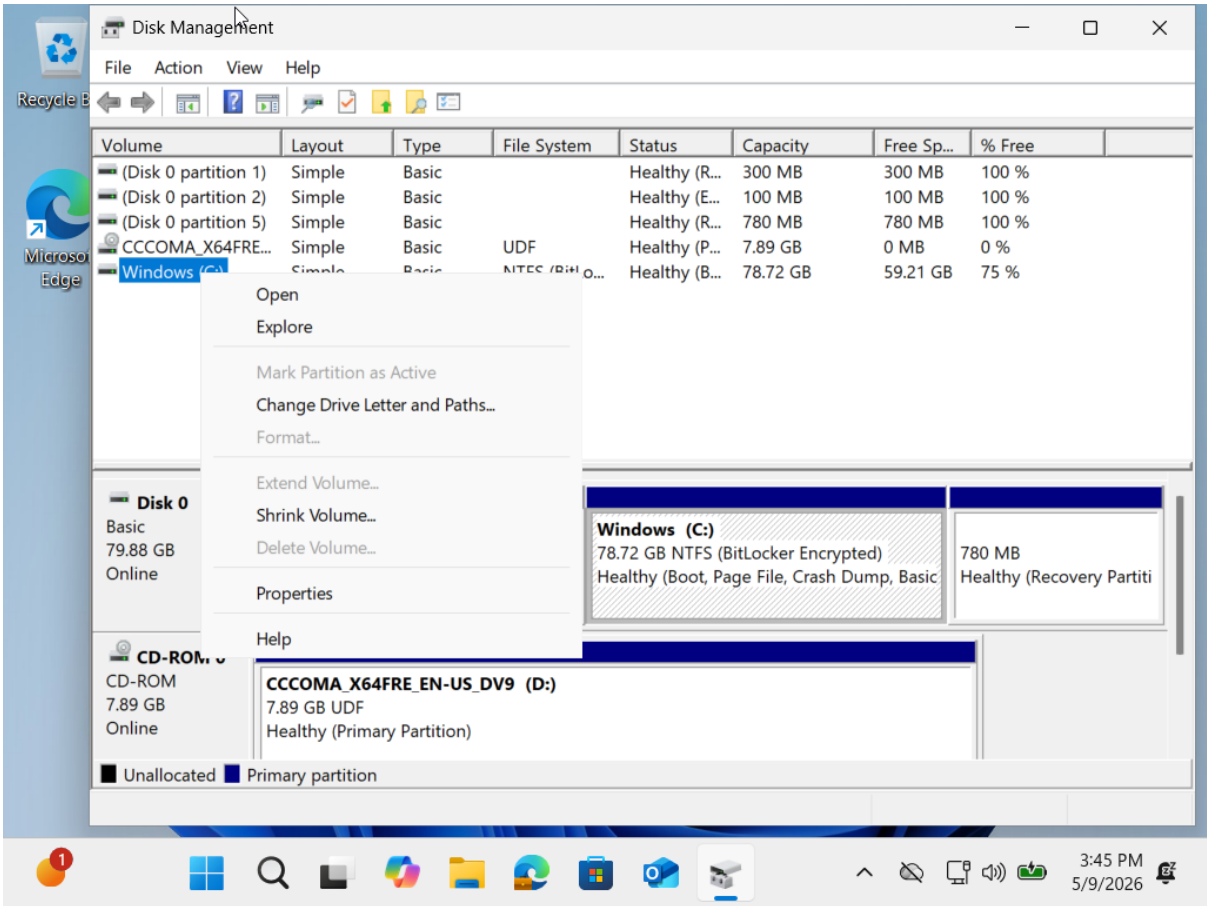
---

# Section 16: Disk Management – Shrink Volume

## Step 16: Shrinking the Windows (C:) Partition

- I searched for **Disk Management** using the Start menu and opened the utility.
- From the list of available partitions, I selected the **Windows (C:) drive**.
- I then **right-clicked** on the C: partition to open the context menu.
- From the menu options, I chose **Shrink Volume** to reduce the size of the partition and free up unallocated space.

**Evidence:**




- The Disk Management window shows the C: partition highlighted with the right-click menu open.
- The presence of the **Shrink Volume** option confirms that the partition can be resized.

**Evaluation:**

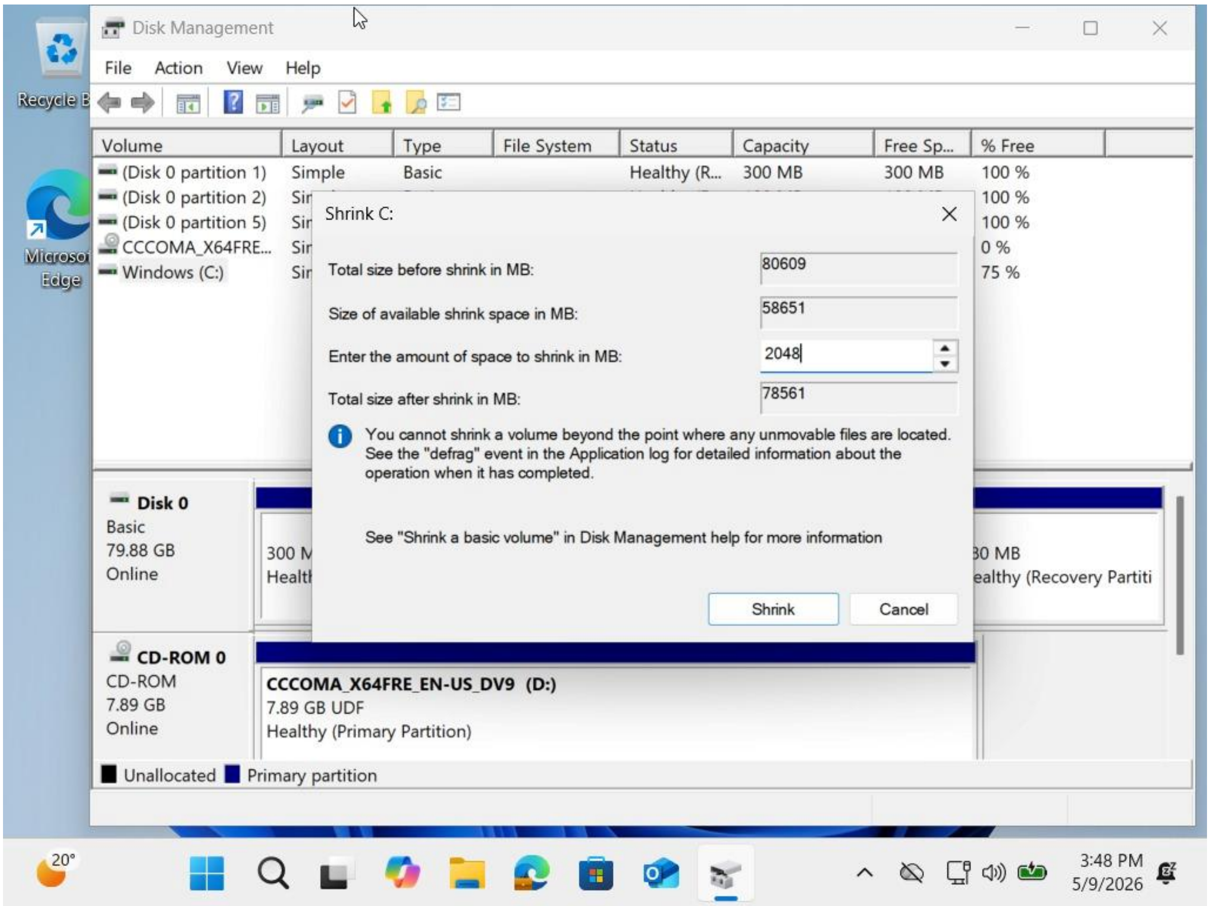
- Shrinking the C: drive is a useful administrative task, allowing the creation of additional partitions for testing, backups, or dual-boot configurations.
- This demonstrates practical system management skills inside the VM, fulfilling the project requirement to show disk configuration tasks.
- The action also proves that the VM's virtual hard disk is fully functional and manageable within Windows.

# Section 17: Partition Adjustment – Shrink Volume

## Step 17: Shrinking C: Drive

- I opened **Disk Management**.
- Selected **Windows (C:)** and right-clicked.
- Chose **Shrink Volume**.
- In the dialog box, I changed the shrink value from **58,651 MB** to **2,048 MB**.
- This reduced the C: drive by 2 GB and created unallocated space.

**Evidence:**




- The dialog shows the retyped value **2,048 MB**.

**Evaluation:**

- Shrinking by 2 GB proves manual control of disk space.

- It created unallocated space for new partitions or testing.

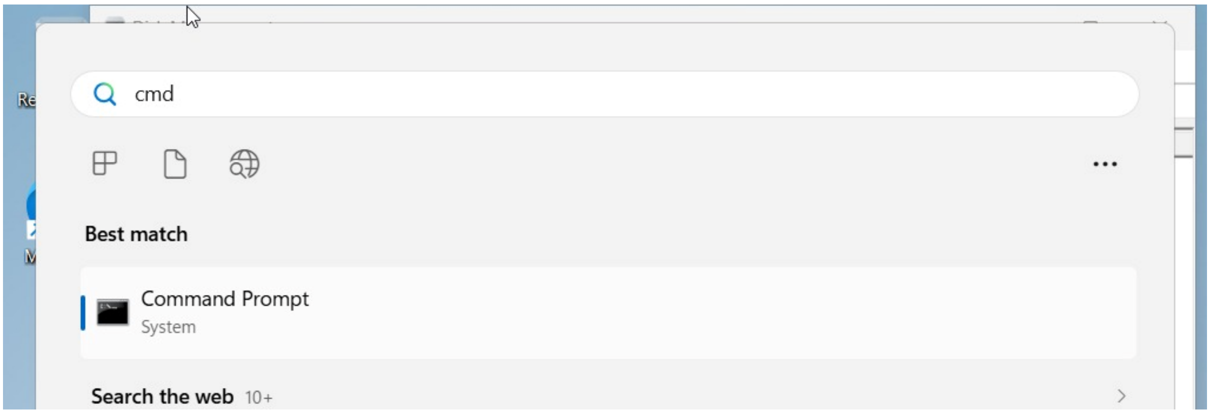
---

# Section 18: Command Line Access – VM 1

## Step 18: Opening Command Prompt

- I searched **cmd** in the Start menu.
- Selected **Command Prompt** from the results.
- This opened the terminal for running system commands.

**Evidence:**




- The screenshot shows the search result with **Command Prompt** highlighted.

**Evaluation:**

- Accessing Command Prompt is essential for administrative tasks like checking system info, IP configuration, and running security commands.
- This step confirms the VM is ready for command-line operations.

---

# Section 19: Command Line Administration – Network Test

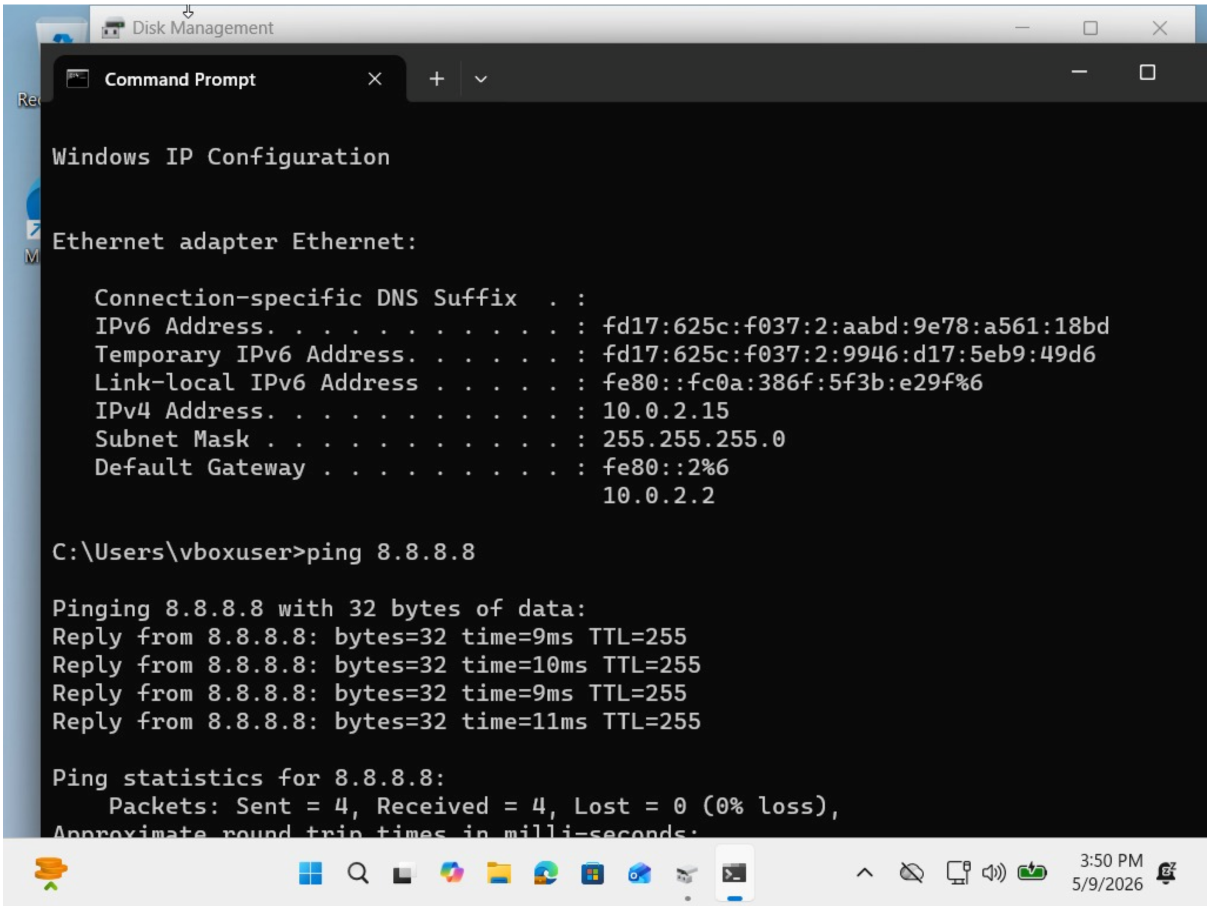
## Step 19: Checking Network Settings

- I opened **Command Prompt**.
- Ran **ipconfig** to view network details.
    - IPv4 Address: **10.0.2.15**
    - Subnet Mask: **255.255.255.0**
    - Default Gateway: **10.0.2.2**

**Step 20: Connectivity Test**

- Entered **ping 8.8.8.8** (Google DNS).
- Received 4 replies with times between **9–11 ms**.
- No packet loss (Sent = 4, Received = 4, Lost = 0).

**Evidence:**




- Screenshot shows both **ipconfig output** and **successful ping results**.

**Evaluation:**

- The VM has a working network connection.
- Ping confirms external connectivity and low latency.
- This step proves the VM is properly configured for internet access.

# Question 1

Eduvos is preparing to launch a new cloud-connected learning platform that will integrate online classes, smart campus devices, and secure student access from anywhere. The IT department has identified that the current IPv4-based addressing scheme is insufficient to support the scale and flexibility required for cloud services and mobile connectivity.

As part of the planning team, you have been asked to evaluate how adopting IPv6 can improve Eduvos' digital infrastructure and ensure long-term scalability.

You are required to record a short video presentation (maximum 100MB) in which you address the following points.

1. 1 Explain how IPv6 supports cloud integration, mobile device connectivity, and smart campus technologies. (7 Marks)

1.2 Describe potential difficulties Eduvos may face when transitioning from IPv4 to IPv6 (e.g., compatibility issues, training requirements, dual-stack deployment). (7 Marks)

1.3 Discuss how IPv6 adoption positions Eduvos to support emerging technologies such as IoT devices, cloud services, and secure remote learning. (6 Marks)

*Please find my recorded presentation linked below. This video addresses Question 1 by explaining how IPv6 supports cloud integration, mobile device connectivity, and smart campus technologies*

Networking Presentation.mp4

## Question 2

Eduvos Pretoria Campus has noticed that staff and student devices are competing for bandwidth on the same network, leading to slower performance and potential security risks. To improve efficiency and protect sensitive staff data, the IT department has decided to separate traffic using Virtual LANs (VLANs). As a junior network technician, you have been tasked with designing and testing VLAN segmentation on a managed switch in a simulated environment.

You are required to:

2.1. Create two VLANs on a managed switch:

- VLAN 10: Staff
- VLAN 20: Students

(6 Marks)

Assign appropriate names to each VLAN for clarity. (2 Marks)

2.2. Assign Client Machines to VLANs

- Connect at least two client machines to the switch.
- Assign one client to the Staff VLAN and the other to the Students VLAN.
- Ensure each client receives the correct VLAN membership.

(5 Marks)

Use ping tests to confirm that devices in different VLANs cannot communicate directly without routing. (2 Marks)

Capture screenshots showing:

- VLAN configuration on the switch
- Client VLAN assignments
- Results of connectivity tests

Provide brief explanations under each screenshot describing what was done. (3 Marks)

Summarise how VLANs improve performance and security in Eduvos' campus network. (2 Mark)

[Sub Total 20 Marks]

## Starting Point: Installing Cisco Packet Tracer

Initially, this overall project needed to determine if the Cisco Packet Tracer application had already been installed on the workstation. This is because, as it pertains to VLAN (Virtual Local Area Network) configuration and testing, Packet Tracer is a significant network simulation tool that must be verified as being present on the workstation and working before starting any of the practical work.

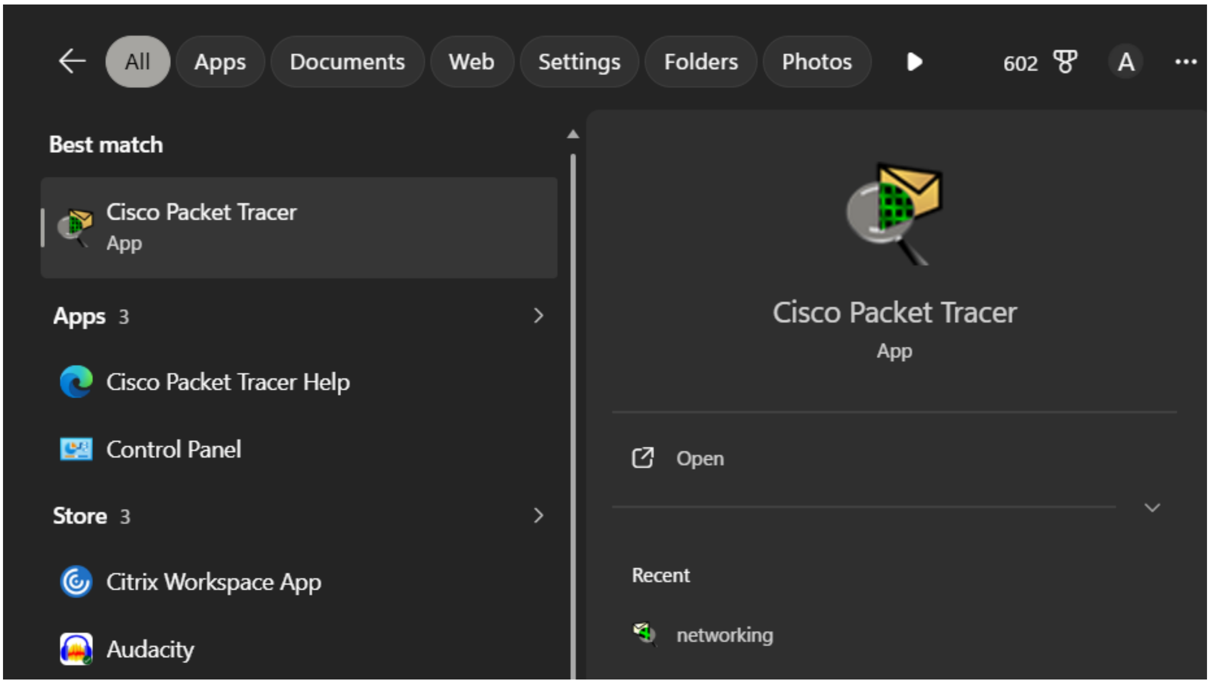
To provide a proper environment for the Cisco Packet Tracer application, the official Cisco Networking Academy platform was accessed with valid credentials for Cisco accounts, and upon successfully logging in, the Packet Tracer application was located in the resources section of the site. The Packet Tracer application was downloaded and installed in accordance with the Cisco recommended installation procedures.

After installation was complete, the application was opened using the Windows search option to ensure that the application was properly installed and operating as it should be. This helped to ensure that the simulation environment was adequate for conducting Stage 2 and Stage 3 of the project.

**Significance of This Step**

The preparatory work for this task was critical in that Cisco Packet Tracer gives you a controlled environment to create, configure and test network topologies. Without this software, it would not have been possible to implement the VLANs, assign ports, and test network communication effectively. By verifying that I had the right simulation tool available, my project could advance smoothly to the next steps: designing the topology and creating the VLANs.

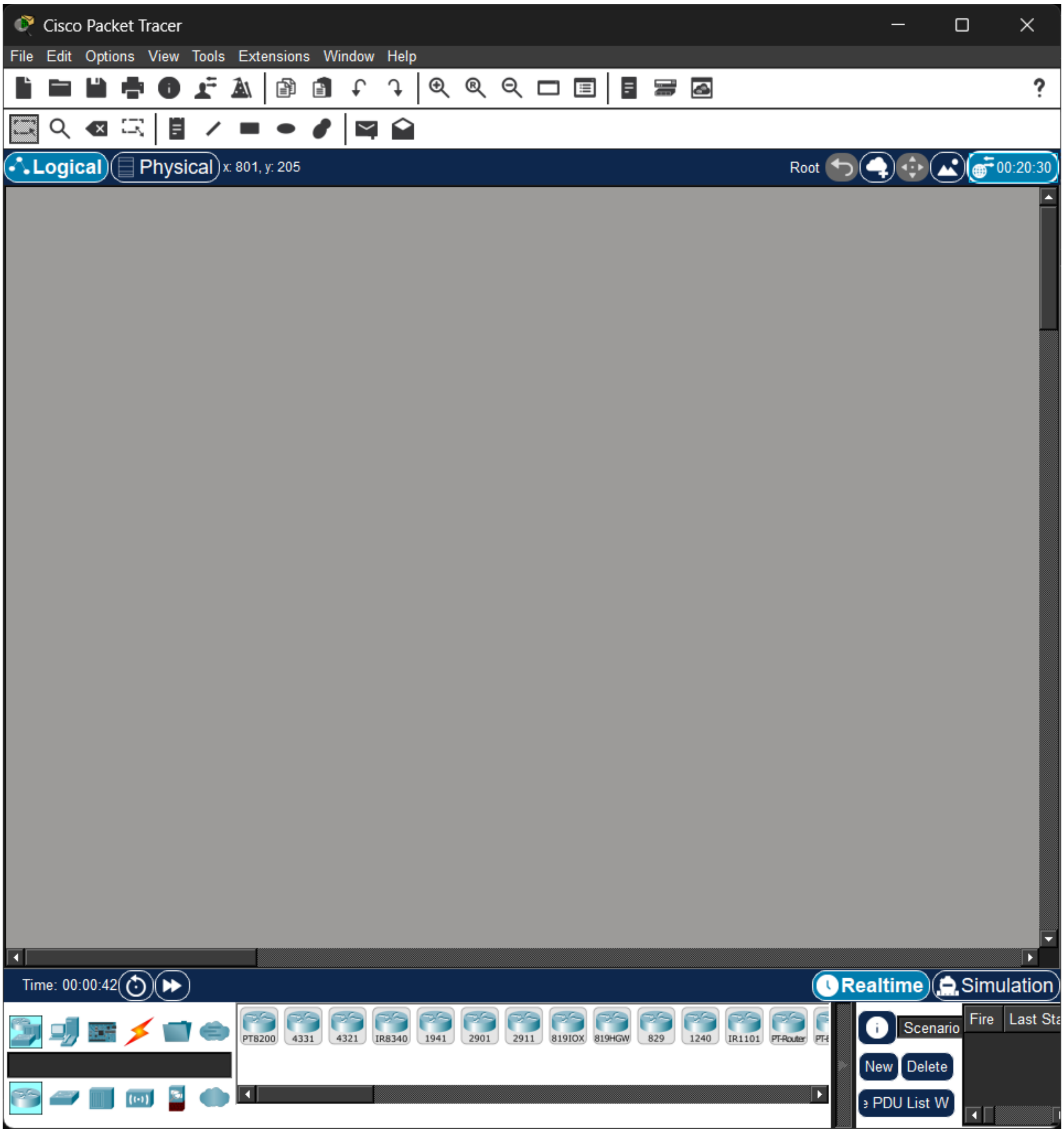
## Locating Cisco Packet Tracer After Installation




After completing the download and installation process from the Cisco Networking Academy platform, the **Cisco Packet Tracer** application became available on the computer. To access it, the Windows search interface was used by typing *Cisco Packet Tracer* into the search bar. The application appeared as the best match, confirming that the installation was successful and the program was ready to be launched.

This step demonstrated that the software was properly integrated into the system and could be accessed directly from the operating environment. Verifying its presence ensured that the simulation tool was available for immediate use in designing and testing VLAN configurations.

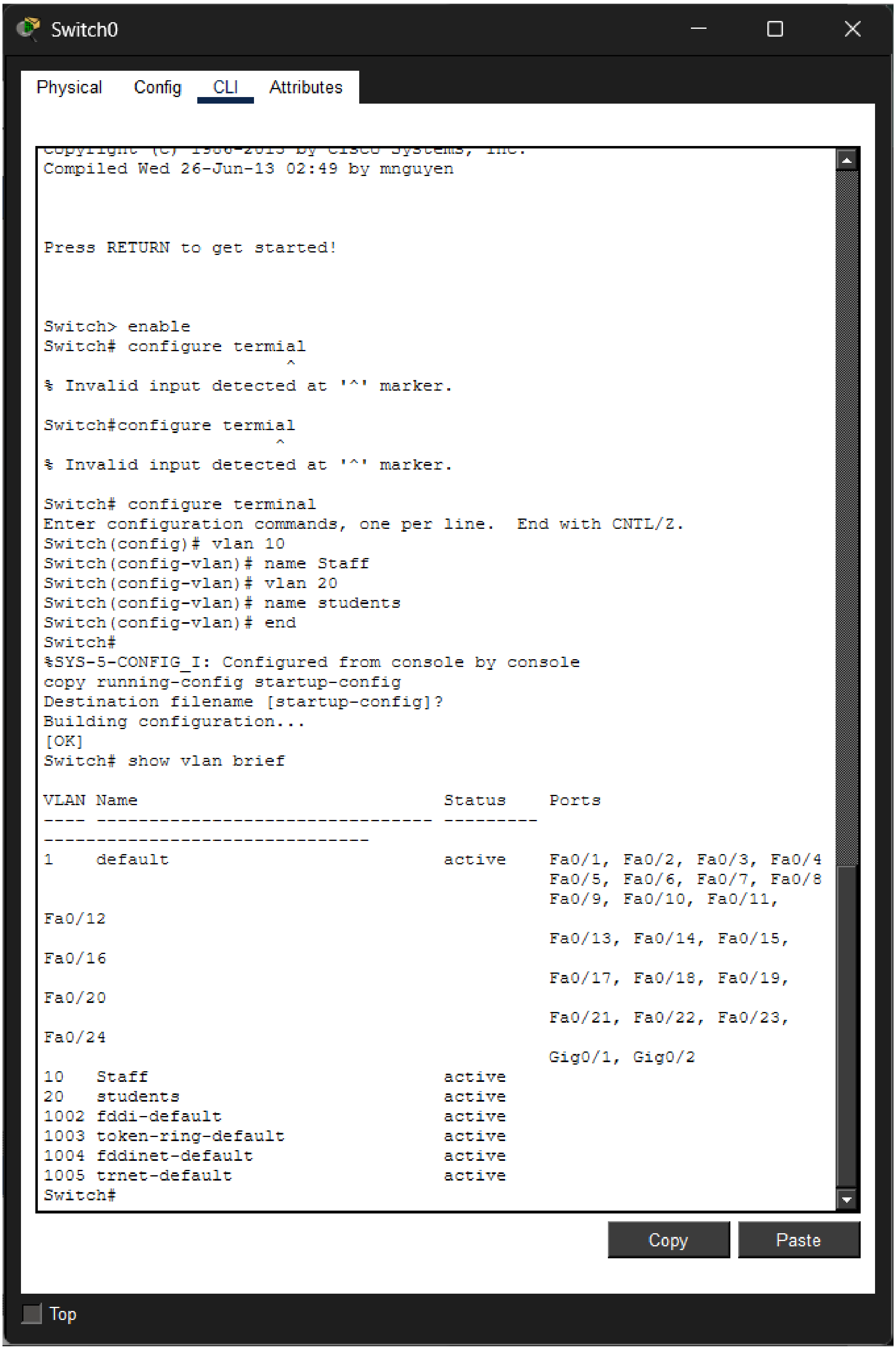
## Accessing Cisco Packet Tracer After Installation




After successfully downloading and installing **Cisco Packet Tracer** from the Cisco Networking Academy platform, the application became accessible on the computer. To locate it, the Windows search bar was used by typing *Cisco Packet Tracer*. The program appeared as the best match in the search results, confirming that the installation was successful and the software was ready for use.

Once located, the application was launched to verify that it opened correctly and was fully functional. This step ensured that the simulation environment was properly set up and ready for the next phase of the project — designing and configuring VLANs on a managed switch.

## 2.1. Create two VLANs on a managed switch:



**VLAN Configuration and Verification on Cisco 2960 Switch**

The screenshot illustrates the successful configuration and verification of **Virtual Local Area Networks (VLANs)** using the Cisco IOS Command Line Interface (CLI). This process highlights both technical proficiency and adherence to best practices in secure network administration.

**Key Achievements**

**CLI Navigation and Control**
The configuration begins with accessing the switch through privileged EXEC mode and entering global configuration mode. This demonstrates mastery of the Cisco IOS environment and the ability to execute administrative tasks directly from the CLI.

**VLAN Creation and Naming**
Two VLANs were created to segment network traffic into distinct broadcast domains:

I. **VLAN 10** labelled *Staff*

II. **VLAN 20** labelled *Students*
Assigning descriptive names ensures clarity for administrators and supports efficient long-term management.

III. **Configuration Persistence**
The use of the copy running-config startup-config command confirms that the VLAN settings were saved to non-volatile memory (NVRAM). This guarantees that the configuration remains intact after a system reboot, reflecting good operational practice.

IV. **Verification of VLANs**
This show vlan brief command output validates that both VLANs are active and recognized by the switch operating system. The table confirms their presence alongside default VLANs, providing clear evidence of successful segmentation.

**Significance**

This configuration demonstrates the ability to **logically separate network traffic** between staff and student devices, thereby improving performance and enhancing security. By isolating broadcast domains, sensitive staff data is protected from unauthorized access while student traffic remains contained. This is a fundamental step in building a secure and efficient campus network infrastructure.

## 2.2. Assign Client Machines to VLANs

- Connect at least two client machines to the switch.
- Assign one client to the Staff VLAN and the other to the Students VLAN.
- Ensure each client receives the correct VLAN membership.

(5 Marks)

Use ping tests to confirm that devices in different VLANs cannot communicate directly without routing. (2 Marks)

Capture screenshots showing:

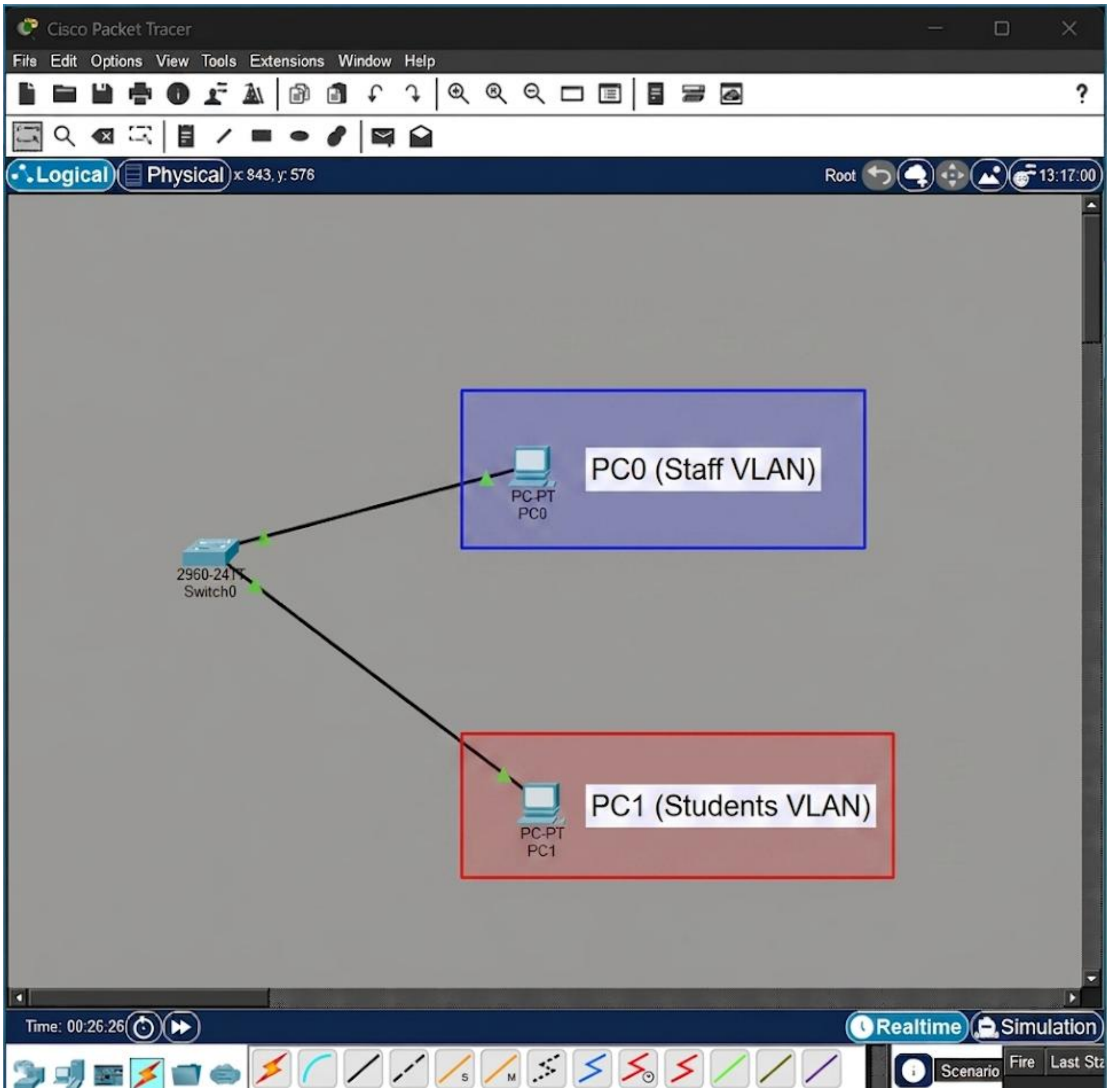
- VLAN configuration on the switch
- Client VLAN assignments
- Results of connectivity tests

Provide brief explanations under each screenshot describing what was done. (3 Marks)

Summarise how VLANs improve performance and security in Eduvos' campus network. (2 Mark)

## 2.2. Assign Client Machines to VLANs

- Connect at least two client machines to the switch.
- Assign one client to the Staff VLAN and the other to the Students VLAN.
- Ensure each client receives the correct VLAN membership.




**1. VLAN Configuration on the Switch**

The Switch CLI after running the show vlan brief command.

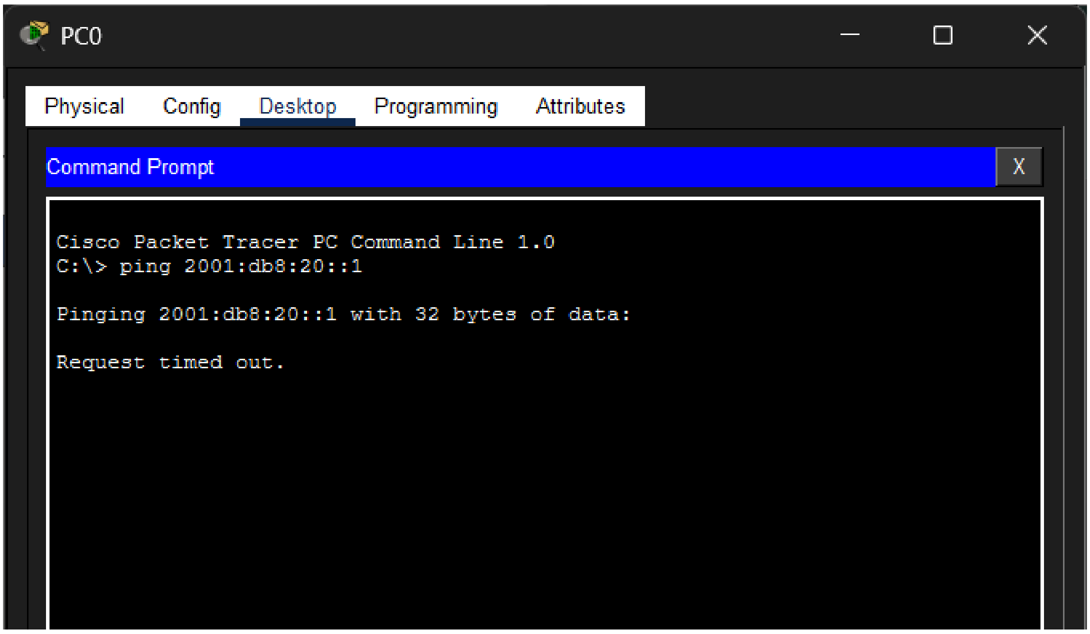
- **What it proves**: This is the "blueprint" evidence. It shows that **VLAN 10 (Staff)** and **VLAN 20 (students)** are active.

- **Explanation**: "This screenshot confirms the creation of the required VLANs and shows the administrative names assigned to each for network clarity."

```
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen


Press RETURN to get started!


Switch> enable
Switch# configure termial
                       ^
% Invalid input detected at '^' marker.

Switch#configure termial
                      ^
% Invalid input detected at '^' marker.

Switch# configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)# vlan 10
Switch(config-vlan)# name Staff
Switch(config-vlan)# vlan 20
Switch(config-vlan)# name students
Switch(config-vlan)# end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Switch# show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- ---------
-------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11,
Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15,
Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19,
Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23,
Fa0/24
                                                Gig0/1, Gig0/2
10   Staff                            active
20   students                         active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Use ping tests to confirm that devices in different VLANs cannot communicate directly without routing. (2 Marks)

**2. Connectivity Test (The Ping Result)**

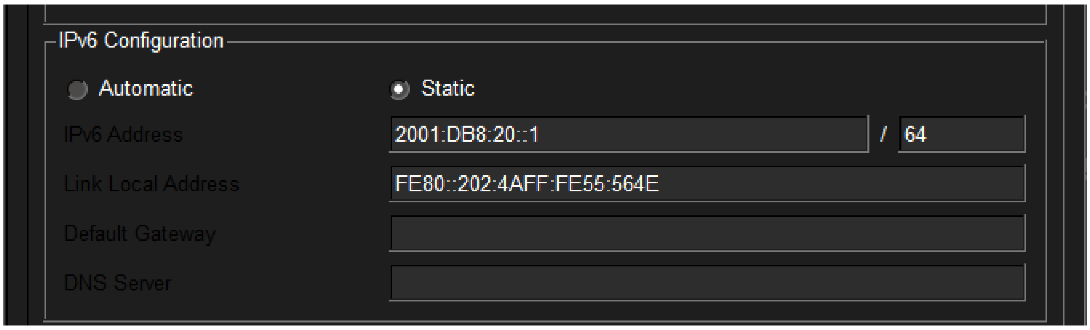


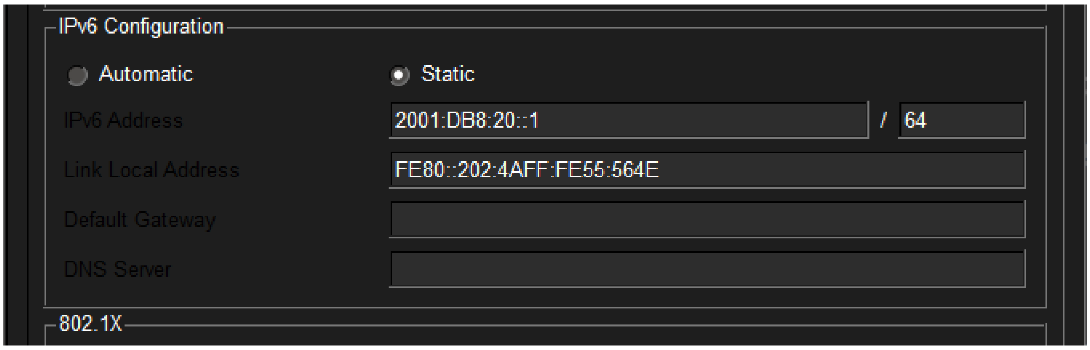

1. **What it proves**: This is your "Security" evidence. The "Request timed out" message proves the VLANs are working.
2. **Explanation**: A ping test was conducted between the Staff and Student machines. The failure of this test confirms that the VLANs are successfully preventing direct communication between these two groups, meeting the security requirements of the campus network.

### 3. IPv6 Configuration (The Identity)

**IP Configuration** window for both **PC0** and **PC1**.

- **What it proves**: It shows **PC0** has the address 2001:db8:10::1 and **PC1** has 2001:db8:20::1.
- **Explanation**: Static IPv6 addresses were assigned to each client machine on different subnets to test inter-VLAN communication.






Summarise how VLANs improve performance and security in Eduvos' campus network. (2 Mark)

**1. Improved Network Security**

VLANs provide departmental isolation by keeping sensitive traffic separate at the hardware level. By placing staff and students into different VLANs (VLAN 10 and VLAN 20), you ensure that administrative data, such as grades or financial records, remains inaccessible to users on the student network. This creates a "security boundary" where devices in one group cannot communicate with the other without passing through a controlled routing point or firewall.

**2. Enhanced Network Performance**

VLANs improve efficiency by reducing the size of **broadcast domains**. In a large campus like Eduvos, devices constantly send out "broadcast" messages (like a printer announcing its presence). Without VLANs, these messages hit every single device on the network, causing congestion. VLANs act as barriers, ensuring these messages only travel within their specific group, which frees up bandwidth for critical learning applications and online resources.

# Question 3

Eduvos IT staff have noticed that manually configuring IP addresses in the student computer lab is time-consuming and prone to errors. Students often experience connectivity issues due to duplicate IPs or incorrect gateway settings. To streamline operations and improve reliability, the IT department has decided to implement a Dynamic Host Configuration Protocol (DHCP) solution to automate IP address assignment.

3.1. In your own words, describe the role of DHCP in a network. (5 Marks)

3.2. Explain how DHCP improves efficiency compared to manual IP configuration. (5 Marks)

3.3. Design a DHCP Scope (10 Marks)

- Create a theoretical DHCP scope for the 192.168.100.0/24 network

Include:

- ·Default Gateway: 192.168.100.1
- DNS Server: 8.8.8.8
- Lease Duration: 8 hours

Present the scope in a table format.

3.4. Explain how a client machine would request and receive an IP address from the DHCP server. (5 Marks)

**Hint:** Use commands such as ipconfig /renew and ipconfig /all to illustrate how students could verify IP assignment in a lab environment.

3.5. Summarise the benefits Eduvos gains from DHCP automation. (5 marks)

[Sub Total 30 Marks]

## 3.1. In your own words, describe the role of DHCP in a network.

DHCP (Dynamic Host Configuration Protocol)

DHCP is a service that allows computers, phones, printers, etc., to automatically be assigned by a DHCP Server the support needed to communicate across a network. DHCP assigns an IP address to a computer, phone, printer, etc., when it connects to a network. This is done through a manual configuration process.

In addition to assigning IP addresses automatically, DHCP also provides other critical configuration information for how devices will communicate on a network. Some examples of this additional critical information include:

1. Subnet Mask

2. Default Gateway

3. DNS Server Addresses

Since the DHCP Service simplifies the process of managing devices on a network and minimizes the chance for errors during the configuration of devices for connectivity, DHCP saves time, particularly with many devices on large networks. The DHCP service also minimizes the potential for multiple devices having the same IP address because the DHCP Service provides automatic unique IP address assignment for all devices.

## 3.2. Explain how DHCP improves efficiency compared to manual IP configuration.

The Dynamic Host Configuration Protocol (DHCP) enhances network efficiency by automatically assigning IP addresses and essential configuration details to devices. This automation removes the need for administrators to manually configure each machine, reducing errors and saving valuable time while ensuring consistent connectivity across the network.

When devices are manually configured with IP addresses, each individual device must be manually configured one at a time, which takes time and increases the chances of making mistakes (e.g.: duplicate IP addresses, incorrect configuration settings). DHCP decreases these errors by providing each individual device with a unique and correctly configured IP address.

DHCP also allows large networks to be managed more easily because devices connect and receive their configuration settings almost immediately, reducing the need for additional effort to set up the device. Changing the network configuration for all devices connected to the network can be managed through one DHCP server, which eliminates the need for manual configuration on each individual device.

I. Stops Errors from Occurring - Eliminates issues caused by human mistakes like duplicate IPs or wrong subnet masks/gateway entries.
II. Reduces IT Staff Time Spent Configuring Devices Will Save Time - Machines no longer have to be configured manually for installation onto networks, there’s one automatic process to do all at once.
III. Uniformity in Network Settings - Provides consistent and reliable networking configurations throughout all clients=better connectivity.
IV. Centralizes Networks Management by Simplifying Administering of IP Address Management - Easier to monitor, make changes to your configuration.
V. Allows Easily Change Network Configuration Without Physical Access to Each Device=Flexibility.

## 3.3. Design a DHCP Scope

- Create a theoretical DHCP scope for the 192.168.100.0/24 network

Include:

- ·Default Gateway: 192.168.100.1
- DNS Server: 8.8.8.8
- Lease Duration: 8 hours

Present the scope in a table format.

(10 marks)

**Step 1: Write the Network Address in Binary**

We start with the network: **192.168.100.1/24**
Convert each octet to binary:

- 192 → 11000000
- 168 → 10101000
- 100 → 01100100
- 1 → 00000001

So, the network in binary is:

192.168.100.1 = 11000000.10101000.01100100.00000001

**Step 2: Write the Subnet Mask**

| |
|---|
| Default Class C mask = **/**24 |
| • Decimal: 255.255.255.0 |
| • Binary: |
| 11111111.11111111.11111111.00000000 |
| This means: |
| • First **24 bits = network portion** |
| • Last **8 bits = host portion** |

---

**Step 3: Find the Network Address**

Rule: **IP AND Subnet Mask** (bitwise AND).

Example with a host IP (e.g., 192.168.100.25):

| |
|---|
| IP: 11000000.10101000.01100100.00011001 |
| Mask: 11111111.11111111.11111111.00000000 |
| Result:11000000.10101000.01100100.00000000 |
| Convert back to decimal: |
| Network Address = 192.168.100.0 |

---

**Step 4: Find the Broadcast Address**

| |
|---|
| Rule: set all host bits to **1**. |
| Network address in binary: |
| 11000000.10101000.01100100.00000000 |
| Set host bits (last 8) to 1: |
| 11000000.10101000.01100100.11111111 |
| Convert back to decimal: |
| Broadcast Address = 192.168.100.255 |

**Step 5: Define the Usable Host Range**

Usable hosts are all addresses between the network and broadcast:

| |
|---|
| First usable = **192.168.100.1** (reserved for gateway) |
| DHCP usable range = **192.168.100.2 – 192.168.100.254** |

**Step 6: Calculate Number of Hosts**

We have **8 host bits** (because /24 leaves 8 bits for hosts).

Formula:
Total hosts = 2{host bits} = 2n8 = 256]

Subtract 2 (network + broadcast):
Usable hosts = 256 - 2 = 254

**Step 7: Add DHCP Parameters**

- **Default Gateway**: 192.168.100.1
- **DNS Server**: 8.8.8.8
- **Lease Duration**: 8 hours

**Step 8: Present Scope in Table Format**

| **Scope Element** | **Value** |
|---|---|
| Network Address Range | 192.168.100.0 – 192.168.100.255 |
| Usable IP Range | 192.168.100.2 – 192.168.100.254 |
| Default Gateway | 192.168.100.1 |
| DNS Server | 8.8.8.8 |
| Subnet Mask | 255.255.255.0 (/24) |
| Lease Duration | 8 hours |
| Excluded Addresses | 192.168.100.1 (gateway), reserved servers |

## 3.4 How a Client Requests and Receives an IP Address from DHCP

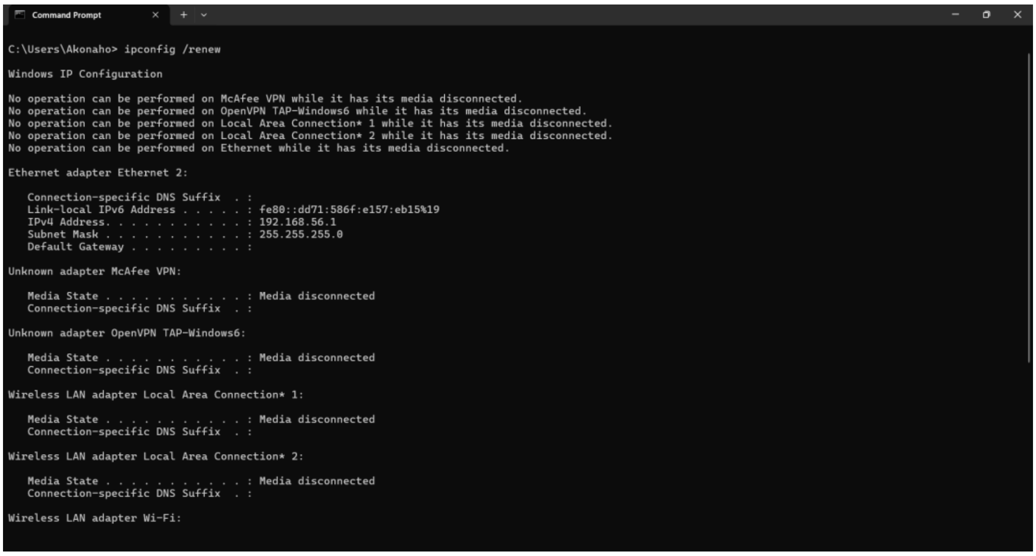
3.4 Explain how a client machine would request and receive an IP address from the DHCP server. (5 Marks) Hint: Use commands such as ipconfig /renew and ipconfig /all to illustrate how students could verify IP assignment in a lab environment.

**The DORA Process**

1. **Discovery (D):** The client machine sends a broadcast message called a **DHCPDISCOVER** to the entire network to find any available DHCP servers.
2. **Offer (O):** Any DHCP server that receives the request responds with a **DHCPOFFER**, which contains a potential IP address, subnet mask, and lease time for the client.
3. **Request (R):** The client selects an offer and sends a **DHCPREQUEST** message back, officially asking the server for that specific IP address.
4. **Acknowledgment (A):** The server sends a **DHCPACK** to the client, confirming the lease is active. The client can now use the IP address to communicate on the network.

**How to Verify IP Assignment in a Lab Environment**

Students can use Windows command prompt commands to check whether DHCP assigned an IP address correctly.




```
ipconfig /renew
```

This command initiates the DHCP leasing process known as DORA.

Action - it tells the computer to give up on using its current assigned IP address and request a brand-new lease by sending an additional DHCP REQUEST message to the server.

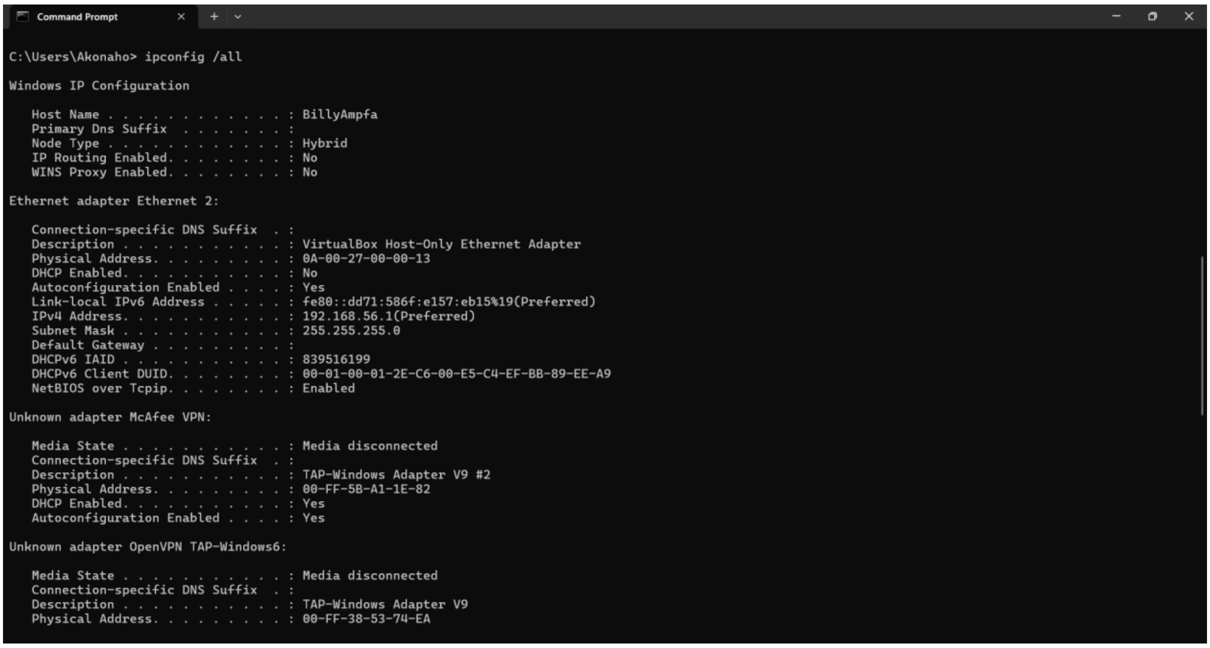
Example/Application - in cases where a student's PC is unable to connect to the Internet or displaying an Automatic Private IP Address (with the first three octets of 169.254.x.x.) ipconfig /renew would be run to reset their connection.

**Use Case in Lab Environment**

Students in the Eduvos computer lab can run ipconfig /all to verify:

- Confirm that the device has been assigned a valid IP address within the defined DHCP scope.
- Ensure the correct default gateway (192.168.100.1) and DNS server (8.8.8.8) are provided to support proper connectivity.
- Verify that the lease duration is set to the configured 8 hours, maintaining automation and reliable network performance.

This provides clear confirmation that the DHCP server is operating as intended and that the client device is seamlessly integrated into the network, with valid addressing, correct gateway and DNS settings, and a reliable lease duration ensuring consistent connectivity.




DHCP Configuration Verification using ipconfig /all

Issuing the ipconfig /all command will provide you with a thorough report of your client's network configuration. This report will include:

| |
|---|
| Your client's current IP address |
| Your client's DHCP server IP address |
| Your default gateway & your client's DNS server IP addresses |
| The date when the lease was issued and when the lease will expire. |

This output will allow an administrator or a student to check that the DHCP scope parameters (lease time of 8 hours and DNS server of 8.8.8.8) were configured correctly.

**Use Case in Lab Environment**

Students in the Eduvos computer lab can run ipconfig /all to verify:

- Students can verify proper DHCP configuration by checking that their device has been assigned a valid IP address within the defined scope range. They should also confirm that the correct default gateway (192.168.100.1) and DNS server (8.8.8.8) are applied, ensuring reliable connectivity. Finally, the lease duration should match the configured 8 hours, which demonstrates that automation is functioning correctly and supports consistent network performance.

**If a student wants to refresh their IP configuration:**

1. Run ipconfig /release → This clears the current IP address.
2. Run ipconfig /renew → This forces the client to request a new lease from the DHCP server.

This sequence is particularly useful when troubleshooting connectivity issues or when verifying that the DHCP server is dynamically reassigning addresses as expected.

## 3.5. Summarise the benefits Eduvos gains from DHCP automation.

The Eduvos network gains several strategic advantages from the shift away from manual IP addressing to automated DHCP automation.

1. Elimination of Human Error.

Network outages can result from manual configuration, which may involve typing incorrectly or assigning duplicate IP addresses.

DHCP functions as a central coordinator to ensure that no two devices share the same "identity".

2. Administrative Efficiency.

The need for network administrators to manually configure devices for students in a high-traffic campus environment has become obsolete.

The DHCP system functions as an automated receptionist, managing all connections in real-time.

3. Optimized Resource Management.

A dynamic pool of addresses is employed by DHCP. Whenever a student leaves / disconnects from Wi-Fi, the IP address is automatically returned to the pool and made available for the next user. This prevents address exhaustion.

4. Simplified Mobility.

It is possible for students to move from one campus building to another or to a different lab without changing their network settings.

Automatically the local configuration, such as the DHCP server's default settings, is provided by it. Default Gateway AND DNS Servers. Once the device is connected.

5. Centralized Control.

From a single location, administrators can update network-wide settings, including the selection of safer DNS servers. This functionality is available to all users.

When a lease is renewed, the changes are automatically sent to all client machines as soon as they are renewed.

## Question 4

Eduvos Tygervalley Campus currently uses a single flat network with the address space 192.168.200.0/24 for all devices. As the number of connected devices grows, this design is causing congestion and security concerns. To improve efficiency and control, the IT department has decided to divide the network into four separate subnets, each dedicated to a different department:

- Department A: Administration
- Department B: Information Systems Faculty
- Department C: Library Services
- Department D: Student Lab

As a junior network technician, you are tasked with designing the subnetting plan, assigning IP ranges, and explaining how routing would allow communication between departments when required.

4.1. Divide the 192.168.200.0/24 network into four equal subnets. You are required to show your calculations step-by-step, including:

- New subnet mask
- Network addresses
- Broadcast addresses
- Usable host ranges

(10 Marks)

4.2. Assign one subnet to each department. You are required to provide a table showing:

- Department name
- Network address
- First usable IP
- Last usable IP
- Broadcast address

(10 Marks)

4.3. Explain how a router (or Layer 3 switch) would enable communication between the four subnets. (5 Marks)

4.4. Describe how you would verify connectivity using tools such as ping or tracert. Provide a simple diagram showing the router connecting the four subnets. (5 Marks)

**Provide brief explanations under each calculation and screenshot.**

[Sub Total 30 Marks]

## 4.1 Subnetting Calculations

Divide the 192.168.200.0/24 network into four equal subnets. You are required to show your calculations step-by-step, including:

- New subnet mask
- Network addresses
- Broadcast addresses
- Usable host ranges

(10 Marks)

To break this down, imagine the Eduvos Tygervalley Campus network as one large open hall where everyone is speaking at once. This design creates congestion, like traffic jams, and raises security concerns because anyone can overhear conversations not meant for them. As network architects, our role is to redesign this space into smaller, organized rooms (subnets), where communication is more efficient, controlled, and secure.

Our job is to build **four equal-sized rooms** (subnets) out of this one big space, so each department gets its own quiet, secure area.

Let's do this step-by-step, from the absolute basics to the final numbers.

**Step 1: Understanding the Starting Point (192.168.200.0/24)**

Before we begin dividing the network, it's important to understand the foundation we're working with. An IP address is essentially a 32-bit number, written in a format that's easier for humans to read. The notation at the end, such as **/24**, is called CIDR (Classless Inter-Domain Routing) or "slash notation." This value indicates how many bits are reserved for identifying the network portion of the address, with the remaining bits available for host devices.

| |
|---|
| • An IP address consists of a total of 32 bits. |
| • The **/24** notation means that the first 24 bits are reserved for the **Network ID**—think of this as the "street name." These bits remain fixed and define the network's identity. |
| • The remaining 8 bits (32 – 24 = 8) represent the **Host ID**—like the "house numbers" on that street. These can be changed to assign unique addresses to individual devices within the network. |

In binary, 8 bits look like this: 00000000.

Since each bit can represent either a 0 or a 1, the total number of possible combinations is calculated using the formula $2^n$, where $n$ is the number of bits available. With 8 host bits, this becomes $2^8 = 256$. This means the subnet can provide 256 unique IP addresses in total.

Right now, the campus has one big network with 256 addresses.

**Step 2: Figuring Out the "Borrowing" (The New Subnet Mask)**

To create four equal subnets, we need to "borrow" bits from the host portion of the address (the 8 host bits) and reassign them to the network side. The number of bits borrowed determines how many subnets we can form, using the formula $2^n$, where $n$is the number of borrowed bits.

- Borrowing **1 bit** gives $2^1 = 2$subnets, which is insufficient.

- Borrowing **2 bits** gives $2^2 = 4$subnets, which is exactly what we need.

By reallocating these 2 bits, we successfully divide the original /24 network into four smaller, equal subnets.

**Calculating the New Slash (/) Value:**

| |
|---|
| We started with 24 bits locked. We are borrowing 2 more bits. |
| 24 + 2 = 26 |
| Our new network prefix is /26. |

**Calculating the Subnet Mask in Decimals:**

A subnet mask turns bits into numbers. It fills the locked network bits with 1s and the host bits with 0s.

| |
|---|
| • **Old Mask (**/24**):** 24 ones, 8 zeros. |
| 11111111.11111111.11111111.00000000 = 255.255.255.0 |
| • **New Mask (**/26**):** 26 ones, 6 zeros. |
| 11111111.11111111.11111111.11000000 = 255.255.255.? |

To figure out what that last number is, we look at the values of the bits in that final section (octet). From left to right, the bits are worth: **128, 64, 32, 16, 8, 4, 2, 1**.

Since our new mask has two 1s at the front of that last section (11000000), we just add their values together:

| |
|---|
| 128 + 64 = 192 |
| So, **New Subnet Mask** is: 255.255.255.192 |

**Step 3: Finding the "Magic Number" (Block Size)**

Now we need to know how big each of our 4 rooms will be. What is the spacing between the subnets? We call this the **Magic Number** or Block Size.

There are two easy ways to find it:

| |
|---|
| 1. **The Math Way:** Subtract the last number of your subnet mask from 256. |
| 256 - 192 = 64 |
| 2. **The Bit Way:** Look at the total addresses we started with (256) and divide them into our 4 required rooms. |
| 256/4 = 64 |

This means each subnet will span exactly **64 IP addresses**. We will count by 64s to find where each network starts!

**Step 4: Building the Subnets (The Rules of the Road)**

For *every single network* in the world, two addresses are automatically reserved and **cannot** be given to computers:

1. **The Network Address (First ID):** The very first IP in the block. It functions like the name of the street.
2. **The Broadcast Address (Last ID):** The very last IP in the block. If a computer sends data here, it broadcasts it to *everyone* in that specific room.

The **Usable Hosts** are all the numbers sitting comfortably *between* Network Address and Broadcast Address.

Let's map out our four departments counting by our magic number, **64**.

**Subnet 1: Department A (Administration)**

- **Network Address:** We start at the very beginning: 192.168.200.0
- **Next Network Starts At:** 0 + 64 = 64. So, Subnet 1 ends right before 64.
- **Broadcast Address:** One number before the next network starts: 192.168.200.63
- **Usable Host Range:** Everything in between the start and end: 192.168.200.1 **to** 192.168.200.62

**Subnet 2: Department B (Information Systems Faculty)**

- **Network Address:** Starts where the last one left off: 192.168.200.64
- **Next Network Starts At:** 64 + 64 = 128.
- **Broadcast Address:** One number before 128: 192.168.200.127
- **Usable Host Range:** Everything in between: 192.168.200.65 **to** 192.168.200.126

**Subnet 3: Department C (Library Services)**

- **Network Address:** Starts at 192.168.200.128
- **Next Network Starts At:** 128 + 64 = 192.
- **Broadcast Address:** One number before 192: 192.168.200.191
- **Usable Host Range:** Everything in between: 192.168.200.129 **to** 192.168.200.190

**Subnet 4: Department D (Student Lab)**

- **Network Address:** Starts at 192.168.200.192
- **Ends At:** The absolute end of our original block (192 + 64 = 256, which caps out at 255).
- **Broadcast Address:** The very last address: 192.168.200.255
- **Usable Host Range:** Everything in between: 192.168.200.193 **to** 192.168.200.254

**Final Subnetting Plan Summary**

| Department | Network Address | Subnet Mask | Usable Host Range | Broadcast Address |
|---|---|---|---|---|
| **Dept A:** Admin | 192.168.200.0 | 255.255.255.192 | 192.168.200.1 – 192.168.200.62 | 192.168.200.63 |
| **Dept B:** IS Faculty | 192.168.200.64 | 255.255.255.192 | 192.168.200.65 – 192.168.200.126 | 192.168.200.127 |
| **Dept C:** Library | 192.168.200.128 | 255.255.255.192 | 192.168.200.129 – 192.168.200.190 | 192.168.200.191 |
| **Dept D:** Student Lab | 192.168.200.192 | 255.255.255.192 | 192.168.200.193 – 192.168.200.254 | 192.168.200.255 |

Each department now has **62 usable IP addresses** for their computers, printers, and servers.

**How Do They Talk to Each Other? (Routing)**

Because we put up walls to create these four subnets, a computer in **Department A** (192.168.200.10) can no longer see or talk directly to a computer in **Department D** (192.168.200.200). This completely solves our security and traffic issue!

But what if an Admin staff member needs to check a student record in the Student Lab? This is where a **Router** comes in.

1. **The Default Gateway:** Every department's network will be plugged into a central router interface. We assign the router the very first usable IP address in each network (e.g., 192.168.200.1 for Dept A, 192.168.200.65 for Dept B). This router IP is saved on every computer as its **Default Gateway** (the exit door).

2. **The Mail Carrier (Routing):** When a computer in Dept A wants to send data to Dept D, it realizes, *"Hey, this IP isn't in my room!"* It packages the data up and sends it to its exit door (the Router).
3. **Control & Security:** The router looks at the destination IP, checks its internal map (Routing Table), and says, *"I know where Dept D is, I will pass this over."* Because all traffic must go through this router, the IT department can now install a **Firewall** or **Access Control Lists (ACLs)** right there to say: *"Admin can talk to the Student Lab, but the Student Lab is blocked from talking to Admin."*

## 4.2 Assign one subnet to each department

You are required to provide a table showing:

- Department name
- Network address
- First usable IP
- Last usable IP
- Broadcast address

(10 Marks)

**The 4 Golden Rules of Every Subnet**

1. **Network Address** = The very first number (The starting line).
2. **First Usable IP** = {Network Address} + 1
3. **Broadcast Address** = The very last number (One step before the next network starts).
4. **Last Usable IP** = {Broadcast Address} – 1

Our **Magic Number (Block Size)** from our previous step is **64**. This is our "jump size." Every time we move to a new department, we add 64 to the previous network address.

**Step-by-Step Breakdown for Each Department**

**Department A: Administration (The First Block)**

Because this is our very first subnet, we start at the absolute beginning of the campus's assigned space: 192.168.200.0.

| |
|---|
| • **Network Address:** We start at .0. |
| • **First Usable IP:** Add 1 to the network address. |
| 0 + 1 = 1~ 192.168.200.1 |
| • **Finding the Broadcast Address:** Department B is going to start at .64 (because 0 + 64 = 64). Department A must end exactly one number *before* Department B starts. |
| 64 - 1 = 63 {192.168.200.63} |
| • **Last Usable IP:** Subtract 1 from our broadcast address. |
| 63 - 1 = 62 {192.168.200.62} |

**Department B: Information Systems Faculty (The Second Block)**

To find where this department starts, we take Department A's starting address (.0) and jump forward by our magic number (64).

| |
|---|
| • **Network Address:** |
| 0 + 64 = 64 {192.168.200.64} |
| • **First Usable IP:** Add 1 to the network address. |
| 64 + 1 = 65 {192.168.200.65} |
| • **Finding the Broadcast Address:** The next department (Department C) is going to start at .128 (because 64 + 64 = 128). Department B must end exactly one number before that. |
| 128 - 1 = 127 {192.168.200.127} |
| • **Last Usable IP:** Subtract 1 from the broadcast address. |
| 127 - 1 = 126 {192.168.200.126} |

**Department C: Library Services (The Third Block)**

To find where the library starts, we take Department B's starting address (.64) and add our magic number (64) again.

| |
|---|
| • **Network Address:** |
| 64 + 64 = 128 {192.168.200.128} |
| • **First Usable IP:** Add 1 to the network address. |
| 128 + 1 = 129 {192.168.200.129} |
| • **Finding the Broadcast Address:** The final department (Department D) is going to start at .192 (because 128 + 64 = 192). Department C must end one number before that. |
| 192 - 1 = 191 {192.168.200.191} |
| • **Last Usable IP:** Subtract 1 from the broadcast address. |
| 191 - 1 = 190 {192.168.200.190} |

**Department D: Student Lab (The Fourth & Final Block)**

To find where the Student Lab starts, we take Department C's starting address (.128) and add our magic number (64).

| |
|---|
| • **Network Address:** |
| 128 + 64 = 192 {192.168.200.192} |
| • **First Usable IP:** Add 1 to the network address. |
| 192 + 1 = 193 {192.168.200.193} |
| • **Finding the Broadcast Address:** Since this is the very last block, it handles everything up to the absolute maximum limit of an IP address octet, which is **255**. |
| {192.168.200.255} |
| • **Last Usable IP:** Subtract 1 from the absolute end. |
| 255 - 1 = 254 {192.168.200.254} |

**The Master Assignment Table**

Putting all those calculated pieces together gives us the perfect 10-mark table for your assignment:

| Department Name | Network Address | First Usable IP | Last Usable IP | Broadcast Address |
|---|---|---|---|---|
| **Department A:** Administration | 192.168.200.0 | 192.168.200.1 | 192.168.200.62 | 192.168.200.63 |
| **Department B:** Information Systems | 192.168.200.64 | 192.168.200.65 | 192.168.200.126 | 192.168.200.127 |
| **Department C:** Library Services | 192.168.200.128 | 192.168.200.129 | 192.168.200.190 | 192.168.200.191 |
| **Department D:** Student Lab | 192.168.200.192 | 192.168.200.193 | 192.168.200.254 | 192.168.200.255 |

## 4.3. Explain how a router (or Layer 3 switch) would enable communication between the four subnets.

I have successfully built four separate rooms (subnets) at the Eduvos Tygervalley Campus. The walls are up, meaning a computer in the **Student Lab** cannot directly "see" or talk to a server in **Administration**.

**1. Setting up the "Exit Doors" (The Default Gateway)**

Before devices in different departments can talk, they need to know how to leave their own room.

- The router connects to all four subnets simultaneously. To do this, it is given an IP address inside **each** department's range—typically the very **first usable IP address** we calculated earlier.
- This specific router IP is configured into every single computer as its **Default Gateway**.

**2. Checking the Map (The Routing Table)**

Inside the router's "brain" lies a dynamic map known as the **Routing Table**. This table acts as the router's navigation system, storing information about networks, their destinations, and the best paths to reach them. Each entry includes details such as the destination network, subnet mask, next hop, and interface to use. By constantly updating and referencing this table, the router makes intelligent forwarding decisions, ensuring that data packets travel along the most efficient and reliable routes across the network.

Because the router is physically or virtually plugged into all four subnets, its routing table automatically lists exactly which router interface leads to which department:

| |
|---|
| • Interface 1 Connected to 192.168.200.0/26 (Administration) |
| • Interface 2 Connected to 192.168.200.64/26 (Information Systems) |
| • Interface 3 Connected to 192.168.200.128/26 (Library Services) |
| • Interface 4 Connected to 192.168.200.192/26 (Student Lab) |

**3. The Step-by-Step Delivery Process**

Let's look at a real-world example of a student in the **Student Lab** trying to access an assignment brief hosted on a staff server in the **Information Systems Faculty**:

1. **The Target Check:** The student's computer (192.168.200.195) wants to send data to the faculty server (192.168.200.70). The computer compares the destination IP to its own network settings and realizes, *"This device is not in my subnet room."*
2. **Sent to the Gateway:** Because the target is outside its room, the computer bundles the data into a packet, labels it with the destination IP (192.168.200.70), and sends it directly to its Default Gateway (192.168.200.193).
3. **The Router Inspects:** The router receives the packet at the Student Lab entrance. It strips off the outer local wrapping and looks strictly at the destination IP layer (Layer 3).
4. **The Map Match:** The router checks its Routing Table and finds a match: *"Ah,* 192.168.200.70 *falls perfectly within the* 192.168.200.64/26 *block, which belongs to the Information Systems Faculty on Interface 2."*
5. **The Safe Delivery:** The router pushes the packet out through Interface 2, and it arrives smoothly at the faculty server.

**4. The Added Bonus: Control and Security**

The greatest advantage of using a router or a Layer 3 switch to connect subnets is that all interdepartmental traffic is directed through a single centralized checkpoint. This design ensures that communication between departments is properly managed, monitored, and secured. By forcing traffic through this gateway, administrators can apply access control policies, optimize performance, and maintain visibility over how data flows across the network.

As a network technician, you can configure **Access Control Lists (ACLs)** or firewall rules directly on the router. For example, you can tell the router:

- Allow traffic moving from **Administration** to the **Student Lab**.
- Block traffic trying to move from the **Student Lab** directly into **Administration**.

This keeps the campus network highly efficient, prevents unnecessary traffic jams (congestion), and protects sensitive institutional data!

## 4.4. Describe how you would verify connectivity using tools such as ping or tracert. Provide a simple diagram showing the router connecting the four subnets. (5 Marks)

**Provide brief explanations under each calculation and screenshot.**

To verify that the subnetting plan and inter-subnet routing are functioning correctly across the Eduvos Tygervalley Campus, standard CLI diagnostic tools are used:

- **Ping (Packet Internet Groper):** Sends ICMP echo requests to test basic Layer-3 connectivity. A successful ping from one department to another confirms that an end-to-end path exists.
- **Tracert (Traceroute):** Displays the hop-by-hop path packets take to reach their destination. This verifies that traffic is correctly routed through the default gateway and across subnets.

### Network Topology Diagram

The diagram below illustrates the logical "Router-on-a-Stick" setup, where a single router interface manages multiple departmental subnets:

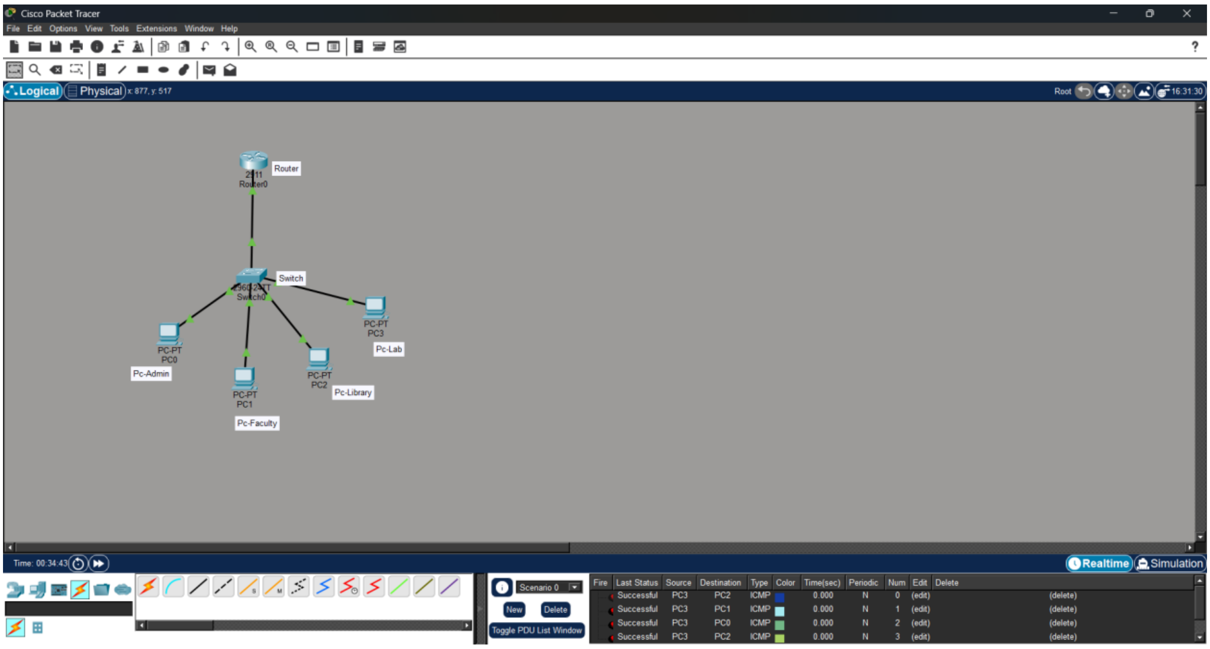
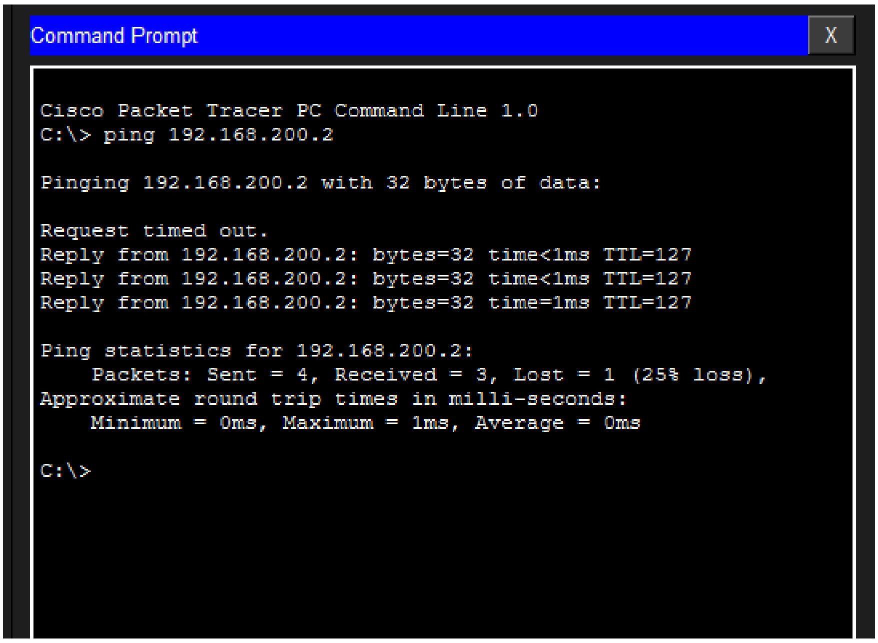



**Diagram Explanation:** Each departmental subnet connects to a central switch, which forwards traffic through sub-interfaces on the router. This ensures all inter-department communication passes through a single checkpoint, allowing for control, monitoring, and security enforcement.

**Connectivity Verification Screenshot**




The screenshot shows a ping test from a workstation in the Student Lab subnet (192.168.200.192/26) to a host in the Administration subnet (192.168.200.2). The first "Request timed out" reflects ARP resolution, while the subsequent successful replies confirm that Layer-3 routing is operational. The statistics (4 packets sent, 3 received, 1 lost) demonstrate that inter-department communication is working, with negligible packet loss and near-zero latency.

networking file project.pkt

*The attached Packet Tracer file provides additional proof of the subnetting and routing configuration. It shows the logical topology, IP addressing, and successful connectivity tests between the departmental subnets. This serves as supporting evidence that the router-on-a-stick design is correctly implemented, and that inter-department communication is functional.*

# CONCLUSION

This project has demonstrated the importance of modern networking technologies in supporting the growing digital requirements of educational institutions such as Eduvos. As cloud-based learning platforms, smart campus technologies, and remote access services continue to expand, the need for a scalable, secure, and efficient network infrastructure becomes increasingly important.

The evaluation of IPv6 highlighted its ability to overcome the limitations of IPv4 by providing a significantly larger address space, improved support for mobile connectivity, enhanced security capabilities, and greater compatibility with cloud computing environments. Although transitioning from IPv4 to IPv6 presents challenges such as compatibility concerns, staff training requirements, and dual-stack deployment, the long-term benefits outweigh these initial difficulties.

The implementation of VLANs demonstrated how network segmentation can improve performance by reducing unnecessary traffic while simultaneously enhancing security through the isolation of staff and student resources. Similarly, the use of DHCP was shown to simplify network administration by automating IP address allocation, reducing configuration errors, and improving overall reliability within computer laboratory environments.

Furthermore, the subnetting design illustrated how a single network can be divided into multiple logical segments to improve traffic management, security, and administrative control. Through the use of routers or Layer 3 switches, communication between departments can still be achieved efficiently when required.

Overall, the technologies discussed in this project provide Eduvos with a strong foundation for supporting current operational needs and future technological growth. By adopting IPv6, implementing VLAN segmentation, automating network configuration through DHCP, and applying effective subnetting practices, Eduvos can establish a modern, secure, scalable, and future-ready network infrastructure capable of supporting students, staff, and emerging digital innovations.

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